Chapter 3

Long-term effects of FTPM, CS and NR on speech

Preface . . . 70
Abstract . . . 77
3.1 Introduction . . . 77
3.2 Methods . . . 83
3.2.1 Participants . . . 83
3.2.2 Stimuli . . . 83
3.2.3 Experimental Procedure . . . 84
3.2.4 Estimation of Frequency-to-Place Factors . . . 85
3.2.5 Data Analysis . . . 86
3.2.6 Data Extraction and Statistical Analysis . . . 88
3.3 Results . . . 90
3.3.1 Overview . . . 90
3.3.2 Study Question 1 - How do Vowel Representations Differ between Groups? . . . 92
3.3.3 Study Question 2 - Do Frequency-to-Place Factors Predict the Displacement of Electrode Confidence Relative to NH Listeners? . . . 97
3.4 Discussion . . . 99
3.4.1 Experienced CI Users Differ Modestly from NH Listeners in Vowel Representation . . . 99
3.4.2 Pre-ECI Users Show Greater Reliance on Apical Electrodes than Post-LCI Users . . . 100
3.4.3 Frequency-to-Place Factors do not Explain the Small Displacement 101
3.4.4 Clinical Implications and Future Perspectives . . . 102
3.4.5 Limitations . . . 103
3.5 Conclusion . . . 103
Postface . . . 104

Preprint - not peer-reviewed

Preface

This chapter presents a preliminary version of the paper in preparation for submission to a peer-reviewed journal: **"Estimation of internal representation of vowels in experienced cochlear implant users and its relationship with frequency-to-place factors"**.

In this preface, we provide an overview of the motivations that led to the test presented here and the pilots undergone to explore the options related to the test.

The primary motivation for this work was to develop a clinical speech-based task capable of capturing the long-term perceptual consequences of frequency-to-place factors in CI users. Such a task could ultimately be used to identify electrodes for which re-mapping might be beneficial. As discussed in the introduction, persistent effects of these factors (*e.g.*, in experienced CI users) are more likely to manifest as degraded sound quality than as reduced speech recognition. We therefore conducted several studies to evaluate sound quality. Three major concerns guided the design of each pilot:

First, we wanted to use **realistic signals** that could adequately represent potential sound quality difficulties faced by CI users in their daily life.

Second, we wanted to develop a task which would be **suitable for clinical use**. In other words, it needed to be time efficient and easily understood by participants.

Finally, we wanted to be able to relate outcomes to frequency-to-place factors so that the perceptual consequences would ultimately be alleviated or managed through re-mapping. In other words, we wanted to **precisely control which electrode was processing the information**.

These three points are reported in the pilots presented below. They offer an overview of how we developed the final experiment presented in this chapter.

Preprint - not peer-reviewed

Pilot 1 - Non-linear FTPM compensation

Approach: To present non-linear frequency shifted versions of realistic speech signals with a non-linear shift designed to exactly compensate each participant's theoretical FTPM profile. Paired comparisons were used with an evolutionary algorithm to converge on the optimal non-linear frequency compensation pattern.

Criteria that pilot was assessed against:

- **Realistic signals:** *Satisfied* - realistic speech signals were used throughout, providing ecological validity for the task.
- **Clinical suitability:** *Not satisfied* - the complex pre-processing pipeline required external software (*i.e.*, MaxMSP), with all stimulus configurations computed upstream before testing. This precluded any real-time adaptation and was impractical in a clinical context.
- **Electrode-level control:** *Not satisfied* - non-linear frequency shifts introduced additional spectral distortions beyond the intended manipulation, and the use of multiple realistic signals made it impossible to determine which part of each signal drove participant decisions. Importantly, the non-linear shifts added or removed spectral information unevenly across conditions, making comparisons unfair at the edges of the shift.

Decision: The non-linear frequency processing concept was promising but overly ambitious for a first pilot, as was the use of an evolutionary algorithm. We therefore decided to reduce the complexity of both the signal processing and the data collection method, and to explore simpler experimental designs.

Pre

Pilot 2 - Linear shift with MUSHRA

Approach: To present linearly frequency-shifted versions of realistic speech signals within a MUSHRA (MUltiple Stimuli with Hidden Reference and Anchor) interface, allowing participants to rate and compare multiple conditions simultaneously.

Criteria that pilot was assessed against:

- **Realistic signals:** *Satisfied* - realistic speech signals were retained, preserving ecological relevance.
- **Clinical suitability:** *Not satisfied* - the MUSHRA paradigm proved too lengthy and cognitively demanding. Furthermore, the notion of a reference signal was poorly suited to our experiment: most participants treated the reference as an implicit upper bound on quality, preventing them from rating any signal above it, even when such a rating might have been appropriate.
- **Electrode-level control:** *Not satisfied* - linear frequency shifting is inherently asymmetric. Depending on the direction of the shift, spectral information at the edges was either lost or added, confounding the intended manipulation with changes in effective bandwidth, and preventing precise targeting of individual electrodes.

Decision: Abandon the MUSHRA paradigm, which was not adapted for clinical use. Moreover, we identified that the notion of sound quality was not always interpreted equally between participants, so we decided to explore the different components underlying sound quality judgements.

Preprint under review

Pilot 3 - Linear shift with paired comparison and different scales

Approach: We presented different linear shifts in paired comparisons and used a Bayesian approach (specifically the Bradley-Terry-Luce method) to classify and extract the shifted configuration with the best sound quality. In the same interface, we included three different rating scales - naturalness, clarity, and overall sound quality - to try to disentangle the different underlying notions of sound quality.

Criteria that pilot was assessed against:

- **Realistic signals:** *Partially satisfied* - realistic signals were still used, but the multi-signal approach revealed that results differed across stimuli, raising the question of which temporal or spectral portion of each signal drove participant decisions.
- **Clinical suitability:** *Partially satisfied* - the paired comparison paradigm was faster and simpler than MUSHRA. However, presenting three rating scales simultaneously - though intended to save time - turned out to be redundant, as participants rated naturalness, clarity, and overall quality consistently, suggesting these scales did not capture independent perceptual dimensions (at least when evaluated simultaneously).
- **Electrode-level control:** *Not satisfied* - as in Pilot 2, linear frequency shifts remained asymmetric and did not permit precise control over which electrode received a given spectral component.

Decision: At this stage, we clearly needed a better understanding of which part of the signal was used by the participant. This was therefore the main motivation for the experiment presented in Chapter 2 (*i.e.*, the relationship between FTPM, sound quality, and sound features). We needed to find a signal whose spectral structure would interact maximally with the different types of FTPM profiles.

Pre-reviewed

Pilot 4 - Band importance frequency

Approach: We needed to understand which part of the signal was important in the sound quality judgement. Different methods exist for this; one of them involves removing one or multiple bands from the signal and observing the effect on sound quality. We decided to do this at the electrode level. However, evaluating all possible combinations can be very long and not appropriate for a clinical assessment. We decided to apply such assessment live using an adaptive Bayesian approach: the idea was to keep the paired comparison approach, present two versions of the signal with different parts of the spectrum present or absent, run a Bradley-Terry-Luce algorithm between each iteration, and target the bands for which no differences were observed in order to compare them in the next trial.

Criteria that pilot was assessed against:

- **Realistic signals:** *Partially satisfied* - realistic signals were still used, and the band-removal approach was a meaningful step towards understanding their spectral structure. However, it became clear that realistic signals were too complex to isolate the contribution of individual electrodes reliably.
- **Clinical suitability:** *Not satisfied* - processing time increased substantially across trials as the number of paired comparisons grew, with considerable delays between trials towards the end. The adaptive targeting strategy used here also proved inefficient: when participants consistently reported no difference between bands, those same bands were repeatedly selected for comparison.
- **Electrode-level control:** *Partially satisfied* - band removal was applied at the electrode level (using the participant mapping), which represented meaningful progress. However, the adaptive algorithm did not converge reliably, limiting our ability to draw electrode-specific conclusions.

Decision: We decided to abandon the use of fully realistic signals at this stage and focused our efforts on less complex signals, for which we could directly study the effect of individual factors at the electrode and frequency level. However, we kept the use of paired comparisons and the Bayesian post-processing due to their efficiency.

Pilot 5a - First vowel experiment

Approach: A task based on vowel perception was developed using natural speech recordings. The second formant frequency (F2) was manipulated continuously between the first and third formant range. This decision was made because the frequency range between F1 and F3 was large enough to apply modifications. For a single vowel, paired comparisons of different F2 positions were presented, and participants selected the version of better perceptual quality using a rating scale. Responses were expected to correlate with individual frequency-to-place factors.

Criteria that pilot was assessed against:

- **Realistic signals:** *Partially satisfied* - moving to isolated vowels represented a meaningful simplification, but the use of real speech recordings still severely constrained the range of manipulations that could be applied without introducing artefacts.
- **Clinical suitability:** *Partially satisfied* - the paired comparison paradigm is fast, simple, and well-suited to Bayesian modelling, which was a clear improvement over earlier approaches. However, the use of natural recordings limited flexibility and reproducibility across participants and sessions.
- **Electrode-level control:** *Not satisfied* - manipulating F2 in a natural vowel inevitably altered vowel identity alongside sound quality, confounding the two perceptual dimensions. The paired comparison paradigm could not disentangle categorical perception effects from genuine quality judgements, and it remained impossible to attribute responses to specific electrodes. Moreover, the range of electrodes studied was limited to the interval between the first and third formant (*i.e.*, where F2 was manipulated).

Decision: The paired comparison paradigm has clear advantages - it is fast, simple, reliable, and well-suited to Bayesian modelling. However, it was not sufficiently adapted to this task. Additionally, the use of recorded speech, even restricted to isolated vowels, severely limited the range of possible signal manipulations.

Proof

These iterative refinements motivated the final experimental design adopted in this chapter. We moved from continuous realistic speech to fully synthetic two-formant vowels, which allowed precise control over the spectral content delivered to individual electrodes while remaining perceptually grounded in speech.

To complement this new stimulus approach, we also replaced the MUSHRA and paired comparison paradigms with a two-stage procedure: participants first identify the vowel, then provide a confidence rating for their identification. This confidence rating was thought to serve as our index of sound quality (or sound clarity). The underlying assumption is that, for a given stimulus, reduced confidence reflects a degraded or unnatural percept rather than genuine between-category confusion - that is, the participant can identify the vowel but does not perceive it as sounding “right”.

We acknowledge that vowel identification in F1/F2 space is inherently difficult, with overlap between vowels even in NH. We therefore do not assume that confidence ratings are a pure measure of sound quality, but rather that they capture a combination of perceptual clarity and categorical uncertainty (even if that latter aspect was partly alleviated through the rejection of responses associated with very low confidence on the rating scale). In the following Chapter, we then preferred to refer as ”internal representation” instead of ”sound quality” to avoid any confusion.

Preprint - not peer-reviewed

Estimation of internal representation of vowels in experienced cochlear implant users and its relationship with frequency-to-place factors

Louis Villejoubert[1], Lorenzo Picinali[2], Kathleen Faulkner[3], Marina Salorio-Corbetto[1], Susan Eitutis[4], and Deborah Vickers[1]

[1]SOUND Lab, Department of Clinical Neurosciences, University of Cambridge, Cambridge, UK
[2]Audio Experience Design, Dyson School of Design Engineering, Imperial College London, London, UK
[3]Oticon A/S, Smørum, Denmark
[4]Emmeline Centre, Cambridge University Hospitals NHS Foundation Trust, Cambridge, UK

Preprint - not peer-reviewed

Abstract

CI users can achieve near-normal speech understanding in quiet, yet many continue to report limitations in sound quality and spectral clarity even after years of device use. The fidelity with which spectral information is conveyed along the electrode array - determined by factors such as frequency-to-place mismatch (FTPM), current spread (CS), and neural responsiveness (NR) - is suspected to contribute to these residual limitations. We developed an original vowel identification and confidence task in which participants selected the most likely vowel from a grid of F1×F2 electrode combinations, providing a continuous confidence rating alongside each response. This allowed us to characterise the optimal spectral representation of each participant and examine whether electrode-level frequency-to-place factors predicted deviations from the NH reference. Because implantation history - whether deafness and CI implantation occurred before or after language acquisition - is known to influence adaptation to the CI signal, two groups of experienced adult users were tested: 10 CI users with congenital onset of deafness, implanted early before language acquisition (Pre-ECI), and 14 CI users with post-lingual onset of deafness, implanted late after language acquisition (Post-LCI), alongside 10 NH listeners as a reference. Results indicated that the vowel representations of CI users were modestly displaced from those of NH listeners overall, with Post-LCI users showing a preference for lower F2 frequencies and Pre-ECI users weighting F1 cues more heavily in their confidence judgements. None of the three frequency-to-place factors accounted for these small deviations at the electrode level, suggesting either that experienced users had fully adapted to these factors at the time of testing, or that the task lacked sufficient sensitivity to detect their residual influence.

3.1 Introduction

Cochlear implantation is the standard of care for restoring auditory function in individuals with severe-to-profound sensorineural hearing loss. Over recent decades, technological improvements, refined surgical techniques, and evolving candidacy criteria have

led to outcomes that were once considered unattainable: the majority of CI users now achieve high levels of speech recognition in quiet after one year of device use (Blamey et al., 2013b; Holden et al., 2013b; Boisvert et al., 2020a).

Despite these encouraging outcomes, perceptual limitations persist beyond the initial adaptation period. One of the most consistently reported residual difficulties concerns sound quality, which many CI users describe as degraded relative to NH even after years of device experience (Caldwell et al., 2017b; Berg et al., 2025a; Dorman et al., 2025). Critically, this degradation is not uniform across recipients: while some users report satisfactory listening experiences across a wide range of everyday situations, others continue to experience substantial perceptual difficulty despite extended implant use (Holden et al., 2013b; Boisvert et al., 2020a). Such inter-individual variability suggests that individual-level factors continue to modulate auditory perception well beyond the initial adaptation period.

One possible source of this variability is the degree of spectral distortion imposed at the level of the ENI, which differs substantially across recipients. CI signal processing maps acoustic frequencies onto a fixed set of electrodes according to a predetermined frequency-to-electrode allocation table. Because the physical position of the electrode array within the cochlea varies across individuals following implantation - determined by insertion depth, array design, surgical approach, and related trauma (Skinner et al., 2002; Noble et al., 2020; Finley et al., 2008) - a systematic mismatch can arise between the frequency a given electrode is intended to represent and the cochlear region it stimulates. The quality of the ENI introduces additional variability: NR and CS differ across electrode sites and across individuals, further shaping the spectral information available to the auditory system and potentially degrading the perceptual quality of individual channels - particularly in cochlear regions affected by neural dead regions (Pfingst et al., 2011a; Bierer, 2007).

The present study focuses on three factors associated with these spectral distortions, collectively referred to here as frequency-to-place factors: FTPM, CS, and NR (for a detailed review of each factor and its known effects on CI perception, see Chapter 1.4). The central question is whether these factors continue to modulate the internal representation of speech in experienced CI users - that is, in individuals who have had sufficient opportunity, due to extensive listening experience, to adapt to the spectral characteristics of their device. As an exploratory extension, we also examine the absolute cochlear position of each electrode - expressed as the characteristic frequency of the cochlear place stimulated (CF_{place}) - to determine whether the physical location of an electrode along the cochlea explains variance in perceptual confidence beyond what is accounted for by its relative FTPM alone. While FTPM captures the deviation between assigned and stimulated frequency, CF_{place} encodes where along the cochlea the electrode resides irrespective of its intended mapping, making the two metrics conceptually distinct despite their mathematical relationship. For example, two electrodes with identical FTPM values (say, -0.5 octaves each) could sit at very different cochlear locations (one apical, one basal), and CF_{place} captures this distinction.

A critical consideration when studying the effect of frequency-to-place factors on speech representation is the listener's auditory history - specifically, listening experience with the CI, duration of deafness prior to implantation, and onset of deafness. We therefore tested two groups. The first group comprised CI users with congenital onset of severe-to-profound deafness, implanted early in childhood before the acquisition of spo-

Preprint - not peer-reviewed

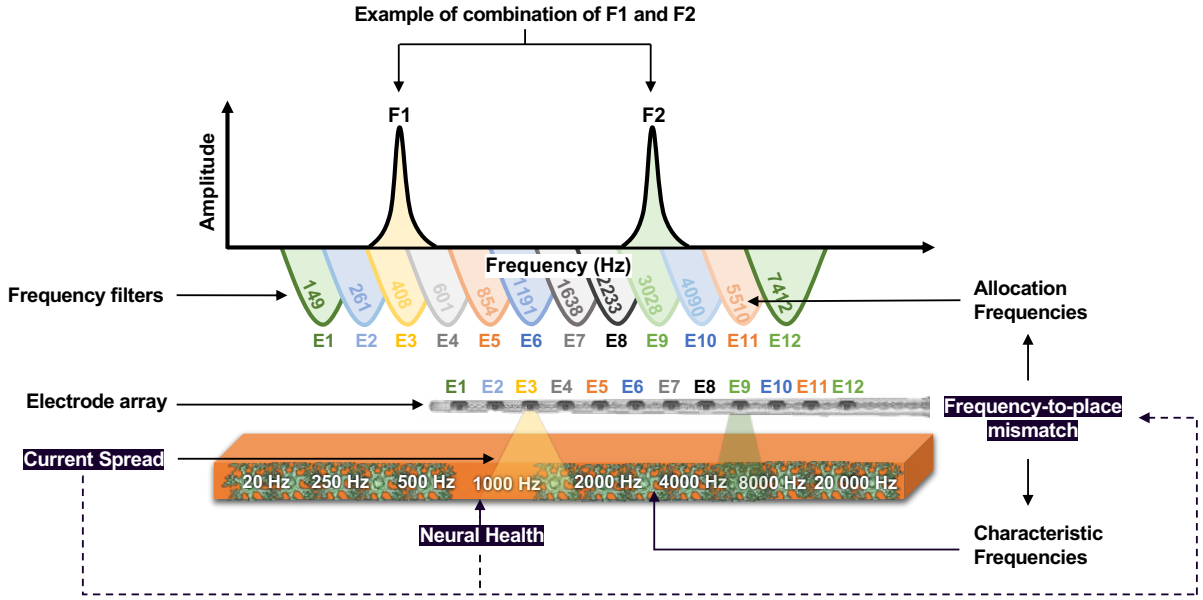


Figure 3.1: **Schematic illustration of the experimental paradigm including the frequency-to-place factors investigated.** The incoming acoustic signal, composed of an F1 and F2 combination, is decomposed into frequency bands and assigned to electrodes E1–E12 as pre-defined by the frequency-to-electrode mapping function, which determines which frequencies are allocated to each electrode. Within the cochlea (bottom), three factors determine the relationship between allocated and perceived frequency. First, electrode array insertion depth sets the baseline FTPM by imposing a shift between characteristic frequencies and allocated frequencies. Second, current spread (coloured triangles) spreads stimulation beyond the target site, recruiting neurons tuned to frequencies outside the frequency band allocated to the electrode being considered; its extent is influenced by local neural health. Third, neural responsiveness varies along the cochlea; where spiral ganglion neurons are absent, the effective site of stimulation may shift to more distant surviving populations. The dotted lines between current spread and neural responsiveness illustrate that these factors are suspected to modulate the primary effect of FTPM.

ken language (Pre-ECI). These individuals developed language and auditory representations exclusively through the CI modality, without ever establishing a natural tonotopic baseline from which to adapt; their internal representation of speech was built entirely around the spectral input provided by their implant, however distorted. The second group comprised CI users with post-lingual onset of deafness, implanted as adults (Post-LCI). These individuals acquired language with natural hearing and subsequently remapped their existing auditory representations onto a spectrally shifted and degraded input once implanted (Peterson et al., 2010; Fu and Galvin, 2007c). These fundamentally different auditory histories may result in different degrees of perceptual adaptation to spectral distortions, and therefore in different relationships between frequency-to-place factors and speech perception representation.

It is worth noting that the speech perception system is inherently flexible: listeners routinely adapt to different speakers' F1 and F2 realisations of the same vowel category, normalising across substantial acoustic variability (Hillenbrand et al., 1995). This flexibility may facilitate adaptation to the shifted formant representations imposed by the CI, though whether it fully compensates for the distortions introduced by frequency-to-place

Preprint - not peer-reviewed

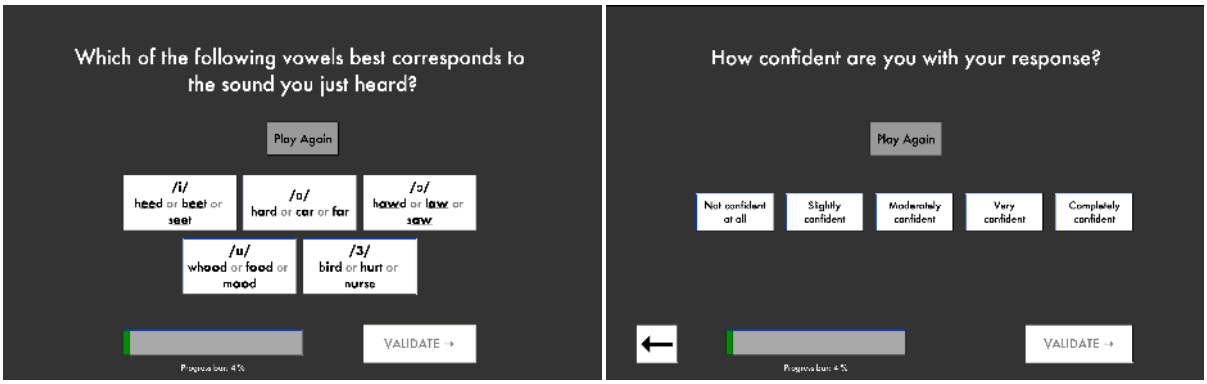


Figure 3.2: **Interface used for the vowel experiment, showing the two response stages.** The left panel shows the first stage, in which the participant selects the perceived vowel from a closed set of five options. The "Play again" button allowed the listener to replay the stimulus. The progress bar indicates the remaining time in the test. Once a vowel is selected, the participant clicks "Validate" to proceed to the second stage. The second stage, shown in the right panel, presents the confidence rating. On a five-point Likert scale ranging from *Not confident at all* to *Completely confident*, participants rated their confidence in their previous identification. They could return to the first stage using the back arrow. Once a confidence rating is selected, they click "Validate" to begin the next trial.

factors in experienced users remains an open question.

The present study examined how the internal representations of vowels differ between NH listeners and experienced CI users - both Pre-ECI and Post-LCI - and how these representations might be shaped by frequency-to-place factors at the individual electrode level.

To probe the internal spectral representation of vowels in experienced CI users, we developed a two-stage vowel identification and confidence rating task. The task exploits the concept of formants - the natural resonant frequencies of the vocal tract, determined by the shape and movement of the vocal organs (lips, teeth, tongue), which produce characteristic spectral peaks in the speech signal. Vowels are defined primarily by the frequencies of their first two formants (F1 and F2) (Peterson and Barney, 1952), and a vowel's position in the F1-F2 space can be summarised by the spectral centroid of each formant region. Because formant frequencies map directly onto locations along the cochlea via the tonotopic organisation of the auditory system, vowel perception provides a natural and ecologically valid probe for place-dependent spectral distortions: any persistent shift, broadening, or degradation of cochlear place coding should be reflected in the responses of CI users relative to their NH peers.

For CI users, stimuli were constructed according to each participant's individual frequency-allocation map, using acoustic stimulation with formants intended to activate two independent channels - one for the F1 frequency region and one for F2.

The key idea of the design is that stimuli are defined in terms of electrode positions rather than fixed acoustic frequencies. Each response can therefore be associated with two specific electrodes, and related to the frequency-to-place factors at those electrodes. For example, if a participant identifies a stimulus activating electrodes E3 and E9 as the vowel [ɑ:] with high confidence, this tells us that the spectral information delivered by

Preprint - not peer-reviewed

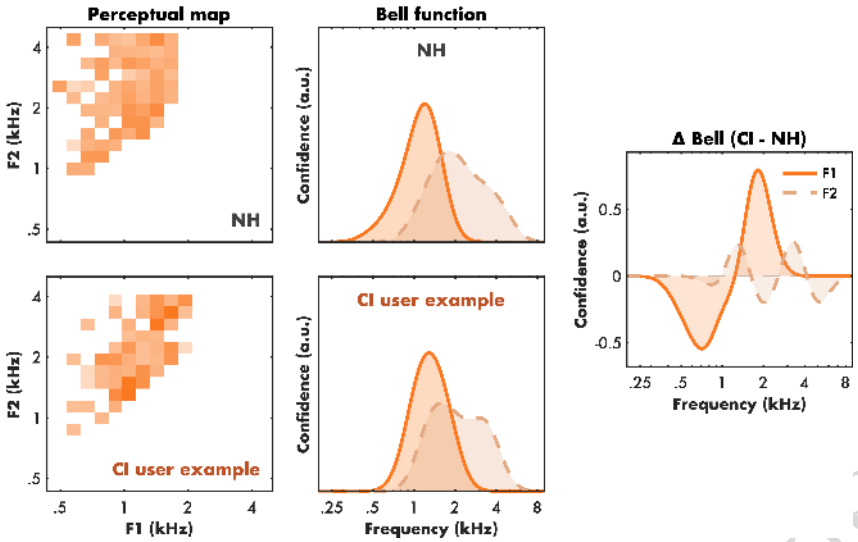


Figure 3.3: **Perceptual map, bell function, and delta bell representation.** The left column shows perceptual maps for vowel [ɑ:] from the NH group (top) and an example CI user (bottom), where colour intensity encodes response confidence across the F1-F2 space. The middle column shows the corresponding bell functions, computed as confidence-weighted kernel density estimates in log-frequency space, with F1 (solid) and F2 (dashed) plotted separately. The right column shows the delta bell (CI − NH), highlighting frequency regions where the CI user perceptual representation deviates from the NH reference.

those two electrode channels is consistent with the internal representation of [ɑ:] for that listener. The design is illustrated in Figure 3.1. For each F1×F2 electrode combination, participants first identified the vowel they perceived from a closed set of five long English vowels (Figure 3.2, left panel), and then rated their confidence of judgement on a five-point scale (Figure 3.2, right panel).

For each participant and each vowel, a Bayesian model was computed to refine the expected confidence response at each F1×F2 electrode combination. The outputs of these models were used to generate perceptual maps across the electrode grid (see left panel of Figure 3.3 for some examples). However, because responses were distributed across multiple competing vowel categories - owing to the perceptual overlap between vowels - some electrode combinations yielded sparse or absent data for a given vowel, even though they would plausibly attract responses under more extensive testing. That is, electrode combinations whose formant frequencies fall within the expected perceptual boundary for a given vowel category may nonetheless receive few responses because participants identified the stimulus as a different, competing vowel on most trials. To address this limitation, perceptual maps were transformed into full distributions of confidence-weighted responses using kernel density estimation, yielding a continuous confidence profile across the electrode array for each participant and each vowel (see middle panel of Figure 3.3). This approach enables a complete comparison along the full acoustical axis rather than at the sparse set of tested electrode combinations.

This primary analytical representation is referred to as the *bell* - which characterises how confidence-weighted perceptual responses are distributed across the acoustical axis for each vowel and participant. The same procedure was applied to NH listeners to construct

Preprint - not peer-reviewed

a group-level NH reference bell, against which individual CI distributions are compared. Three complementary analyses were then applied to this representation.

First, the overall displacement of each CI user bell from the NH reference was quantified using the Wasserstein distance - a measure of distributional dissimilarity that expresses, in octaves, how far a CI participant's confidence distribution is shifted along the frequency axis relative to the NH reference. This provides a single scalar summary of representational displacement per participant, vowel, and formant region.

Second, the direction of that displacement was characterised through a formant shift analysis: the frequency centroid of each participant's bell was compared to the NH reference frequency centroid, yielding a signed shift in octaves where positive values indicate that the CI user's perceptual representation is shifted towards low frequencies relative to the NH reference, and negative values towards high frequencies.

Third, a reliance ratio was computed to quantify the balance of perceptual confidence between the F1 and F2 electrode regions: $R = \text{peak}(\text{bell}_{F2})/(\text{peak}(\text{bell}_{F1})+\text{peak}(\text{bell}_{F2}))$, where values above 0.5 indicate that a participant places greater confidence in their F2 electrode responses than in their F1 electrode responses.

At the individual electrode level, the difference between each CI participant's confidence bell and the corresponding NH reference bell was computed to yield a per-electrode delta (Δ). An example of such difference is provided in Figure 3.3. This delta was then entered as the dependent variable in a model to assess whether the three electrode-neuron interface factors - FTPM, CS, and NR - systematically predicted the deviation of CI confidence profiles from the NH reference. As an exploratory extension, absolute cochlear position (CF_{place}) was additionally included as a predictor.

The task was performed by 14 Post-LCI, 10 Pre-ECI, and 10 NH listeners. NH responses served as reference distributions against which CI users' electrode confidence profiles were compared. Wasserstein distances, frequency shifts, and reliance ratios were computed using the responses of all participants. Frequency-to-place factor analyses were performed on a subset of participants for whom these values were available (see Table 3.1).

The study addressed two main questions:

1. **How do vowel representations differ between groups?** Do experienced CI users - both Pre-ECI and Post-LCI - show systematic differences in their internal vowel representations compared to NH listeners, and do Pre-ECI and Post-LCI differ from each other? This question was addressed through three complementary analyses applied to the electrode confidence bell: the Wasserstein distance from the NH reference (overall displacement), the formant centroid shift (direction of displacement), and the F1/F2 reliance ratio (asymmetry of confidence across the cochlear axis).

2. **Do frequency-to-place factors predict the displacement of electrode confidence relative to NH listeners?** Do FTPM, CS, and NR predict the magnitude and direction of the shift between each CI user confidence profile and the NH reference, and do these relationships differ between Pre-ECI and Post-LCI groups? As an exploratory analysis, does absolute cochlear position (CF_{place}) explain additional variance in this displacement beyond FTPM alone?

Preprint - not peer-reviewed

3.2 Methods

3.2.1 Participants

14 Post-LCI participants (7 female; mean age = 53.7 years, SD = 17.9; mean duration of CI use = 5.9 years, SD = 3.0), 10 Pre-ECI participants (4 female; mean age = 25.8 years, SD = 3.9; mean duration of CI use = 23.4 years, SD = 3.7), and 10 NH listeners (6 female; mean age = 27 years, SD = 4) participated in the study. NH listeners had audiometric thresholds $\leq$ 20 dB HL at octave frequencies from 250 to 8000 Hz bilaterally. Details about CI users are presented in Table 3.1.

CI participants were recruited from the Emmeline Centre for Hearing Implants at Addenbrooke's Hospital (Cambridge, UK). All participants were native speakers of British English. The study was approved by the East of England - Cambridge Central Research Ethics Committee (IRAS number: 335150). All participants gave informed consent before the experiment.

Table 3.1: **Participant demographics and data availability.** Act. el. = number of active electrodes in the clinical MAP. Contra. = contralateral ear condition (HA = hearing aid; CI = cochlear implant; None = unaided). Age at testing is derived from age at implantation plus CI experience. FTPM and PECAP tick marks indicate whether electrode-level frequency-to-place factors were available.

Participant	Sex	Device	Act. el.	Side	Contra.	Age impl.	Age test.	CI exp.	FTPM	PECAP
Pre-ECI 1	M	Cochlear	22	L	HA	2	21	19	✓	✓
Pre-ECI 2	F	Cochlear	20	L	CI	3	19	16	✓	
Pre-ECI 3	F	Cochlear	22	R	HA	2	24	22		
Pre-ECI 4	F	AB	16	R	None	4	29	25		
Pre-ECI 5	M	AB	16	R	CI	1	27	26		
Pre-ECI 6	M	Cochlear	22	L	HA	2	26	24		
Pre-ECI 7	F	Med-El	12	R	HA	2	24	22		
Pre-ECI 8	M	Cochlear	22	L	CI	3	31	28		
Pre-ECI 9	M	Cochlear	22	R	CI	3	30	27		
Pre-ECI 10	M	Cochlear	22	L	HA	2	27	25		
Post-LCI 1	F	Cochlear	22	L	HA	67	72	5	✓	
Post-LCI 2	F	Med-El	12	L	None	45	55	10	✓	
Post-LCI 3	F	Cochlear	22	R	HA	20	24	4	✓	✓
Post-LCI 4	M	Cochlear	22	L	HA	19	22	3	✓	
Post-LCI 5	F	AB	16	R	CI	45	50	5	✓	✓
Post-LCI 6	M	Cochlear	22	L	HA	69	73	4		
Post-LCI 7	M	AB	16	L	None	52	55	3		
Post-LCI 8	F	Cochlear	22	R	None	65	72	7		
Post-LCI 9	M	Med-El	12	R	HA	39	43	4		
Post-LCI 10	F	Cochlear	22	L	HA	21	31	10		✓
Post-LCI 11	F	Cochlear	22	L	CI	67	71	4		✓
Post-LCI 12	M	AB	16	R	None	45	58	13		
Post-LCI 13	M	AB	16	L	HA	65	70	5		
Post-LCI 14	M	Cochlear	22	L	HA	50	56	6		

3.2.2 Stimuli

Synthetic vowel stimuli were created using the Parselmouth library (Jadoul et al., 2018), which provides access to Praat functionality within a Python environment. Each stimulus consisted of a harmonic complex with a fundamental frequency fixed at 80 Hz, associated with a first formant (F1) and a second formant (F2). No higher formants

Preprint - not peer-reviewed

were included. This choice was made to isolate the contribution of F1 and F2 to vowel identification, ensuring that any observed perceptual differences could be attributed to the spectral representation of these two formants rather than to higher-order spectral cues. Each stimulus was 6 s long, with a 500 ms fade-in and fade-out applied at onset and offset. The sampling rate was set at 48 kHz.

NH Stimulus Set

For NH listeners, formant centre frequencies were selected using a Gammatone filter bank. F1 was sampled at 15 frequency values between 234 Hz and 1720 Hz, and F2 at 21 frequency values between 415 Hz and 4093 Hz, with the constraint that F2 was always greater than F1. This yielded 218 unique F1/F2 combinations, each presented four times per participant, for a total of 872 trials.

CI Stimulus Set

For CI users, F1 and F2 values were set to the centre frequencies of the participant's allocated electrodes, as defined by their clinical frequency-allocation map. This ensured that the formants of each stimulus aligned with known electrode stimulation sites, allowing direct comparison between the perceived formant location and the electrode responsible for delivering it. The same constraint as for NH listeners (F2 > F1) was applied. Because the number of active electrodes varied across participants and devices, the number of F1/F2 combinations and therefore the number of trials differed between CI users. Each F1/F2 combination was repeated four times.

3.2.3 Experimental Procedure

The experiment employed a two-stage identification and confidence paradigm. On each trial, a single synthetic vowel stimulus was presented monaurally at a comfortable listening level adjusted individually by each participant.

Stage 1: Vowel identification. Following stimulus presentation, participants identified the vowel they perceived by selecting one of five response options displayed on screen (Figure 3.2, left panel): [ɑː] (as in *start*), [ɜː] (as in *nurse*), [iː] (as in *seat*), [ɔː] (as in *thought*), and [uː] (as in *goose*). These representative words were displayed alongside each vowel symbol to aid identification.

Stage 2: Confidence rating. After selecting a vowel, participants rated their confidence in their choice on a five-point Likert scale ranging from "Not confident at all" to "Very confident" (Figure 3.2, right panel).

Before each testing session, a short training block of 30 trials was conducted to ensure task comprehension. Trials were presented in randomised order. Stimuli were presented using a MacBook Pro, an RME Fireface UCX II sound card, and a Sennheiser HD 600 open-back headset. For CI users, the headset was positioned over their clinical processor.

Preprint – Not Peer-reviewed

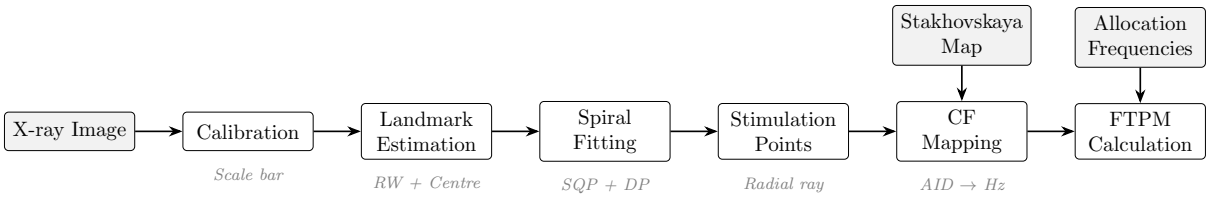


Figure 3.4: **FTPM estimation workflow.** Algorithm for computing electrode-specific frequency-to-place mismatch from post-operative cochlear implant X-rays, combining spiral template fitting with the Stakhovskaya frequency-position function. RW = Round Window, SQP = Sequential Quadratic Programming, DP = Dynamic Programming, AID = Angular Insertion Depth, CF = Characteristic Frequency.

3.2.4 Estimation of Frequency-to-Place Factors

Three electrode-level factors were estimated for a subset of CI participants: FTPM, CS, and NR. Data availability varied across participants, as detailed in Table 3.1. FTPM estimation requires post-operative X-ray imaging; for some individuals, images were either unavailable or of insufficient quality to permit reliable electrode localisation. CS and NR were derived from PECAP recordings, which could not be obtained for all participants due to the absence of a measurable response or device incompatibility. The electrode-level analyses reported in Section 3.2.6 should therefore be interpreted with caution given the reduced and unbalanced availability of these measures.

FTPM Estimation

FTPM was estimated using a semi-automatic algorithm for computing the angular insertion depth of cochlear implant electrodes from post-operative X-ray images (Roberts et al., personal communication). The simplified workflow is shown in Figure 3.4 and representative outputs in Figure 3.5.

After image calibration using the scale bar, electrode contacts were identified manually and key anatomical landmarks - the round window and cochlear centre - were estimated automatically via least-squares circle fitting. A mean cochlear spiral template derived from micro-CT data was then fitted to the electrode positions through 3D rotation optimisation, minimising the sum of squared distances between electrodes and assigned spiral points. For each electrode, the stimulation site was estimated by projecting a line from the cochlear centre through the electrode position to the spiral, and the angular insertion depth relative to the round window (0°) was converted to a CF using the Stakhovskaya spiral ganglion frequency-position function (Stakhovskaya et al., 2007b). The CF at each stimulation site was then compared to the allocated centre frequency of the corresponding electrode to yield the FTPM in octaves for each electrode. The absolute cochlear position of each electrode - expressed as the CF of the place of stimulation (CF_{place}) - was retained from the same estimation pipeline as an exploratory predictor (see Section 3.2.6).

CS Estimation

CS was estimated using the Panoramic ECAP (PECAP) method (Garcia et al., 2021). ECAPs are recorded by delivering a current pulse through a probe electrode and measuring

Preprint - not peer-reviewed

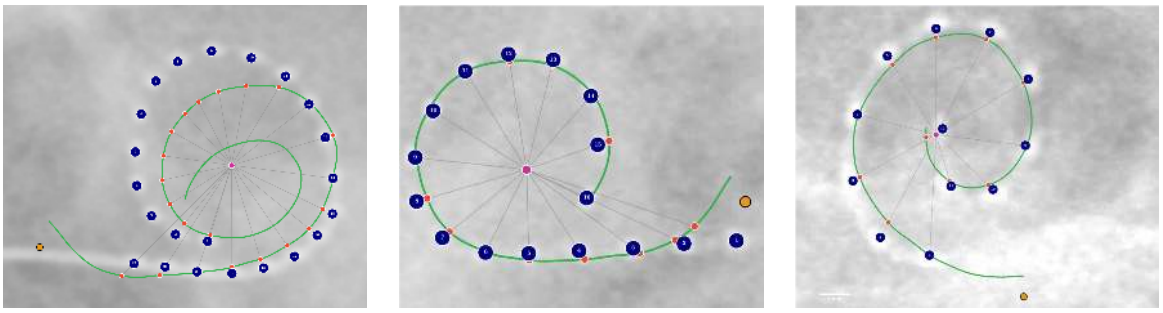


Figure 3.5: **Estimation of FTPM using post-operative X-rays.** Electrodes (blue) are matched to stimulation sites (red) on the fitted cochlear spiral (green) via radial projection from the modiolar axis (magenta). The round window (orange) defines the 0° reference.

the synchronised auditory nerve response at a nearby recording electrode. The PECAP method records ECAPs for all combinations of probe and masker electrodes, yielding an $n \times n$ matrix $\mathbf{M}$. A nonlinear optimisation algorithm then decomposes $\mathbf{M}$ into two vectors: $\boldsymbol{\sigma}$, encoding CS, and $\boldsymbol{\eta}$, encoding NR. Larger $\boldsymbol{\sigma}$ values indicate broader current spread and poorer electrode selectivity. All measurements were recorded at maximum comfortable level (MCL) for each active electrode.

NR Estimation

NR was estimated simultaneously with CS from the same PECAP decomposition (Garcia et al., 2021). The $\boldsymbol{\eta}$ vector encodes the relative responsiveness of auditory nerve fibres along the electrode array, with lower values indicating poorer neural health. Per-electrode $\boldsymbol{\eta}$ values are used as the NR index throughout.

3.2.5 Data Analysis

Exclusion of [iː] Responses

Although five vowel options were presented during the task, responses for the vowel [iː] were excluded from all subsequent analyses (see Section 3.3.1 for justification). All remaining analyses therefore concern four vowel categories: [ɑː], [ɜː], [ɔː], and [uː].

Electrode Confidence Distribution

The main unit computed here is the *electrode confidence distribution* - hereafter referred to as the *bell* - which characterises how confidence-weighted perceptual responses are distributed across the electrode array for each vowel and participant. Rather than projecting responses onto a formant-frequency grid and computing summary statistics in that space, the bell maps the full distribution of responses directly in the acoustical axis, making it directly comparable with the electrode-level factors FTPM, CS, and NR.

Preprint – not peer-reviewed

Electrode Allocation Mapping. For each participant, the CI frequency-allocation map was used to associate each F1/F2 stimulus combination with a pair of electrode positions: the electrode whose allocated centre frequency matched F1, and the electrode whose allocated centre frequency matched F2.

Bayesian Confidence Estimation and Perceptual Maps. For each participant, vowel, and F1/F2 combination, a Bayesian model was fitted to estimate and refine the confidence response. The outputs of these models were used to create perceptual maps in the F1×F2 electrode space.

Confidence-Weighted Kernel Density Estimation. However, responses were sparse in some regions of the perceptual maps: owing to the perceptual overlap between vowels, some combinations yielded little or no data for a given vowel even though they would plausibly attract responses under more extensive testing. To overcome this limitation, perceptual maps were transformed into an aggregated confidence-weighted distribution - the bell - using kernel density estimation (KDE).

For each participant, vowel v, and formant dimension (F1 or F2), responses were aggregated as confidence-weighted counts at each electrode position e:

$$R(e, v) = \sum_{t \in \mathcal{T}(e,v)} c_t \tag{3.1}$$

where $\mathcal{T}(e, v)$ is the set of trials on which vowel v was selected and the corresponding formant stimulus was allocated to electrode e, and $c_t \in \{1, 2, 3, 4, 5\}$ is the confidence rating on trial t. A Gaussian KDE was then applied in log-frequency space over the electrode array to produce a smooth, continuous profile:

$$\text{bell}(f, v) = \frac{1}{Z} \sum_{e} R(e, v) \cdot \exp\left(-\frac{(\log_{10} f - \log_{10} f_e)^2}{2h^2}\right) \tag{3.2}$$

where f is a frequency grid point, f_e is the centre frequency of electrode e, h is the kernel bandwidth (set to 0.10 in $\log_{10}$-Hz units, approximately one-third of an octave), and Z is a normalisation constant ensuring that the bell integrates to unity over the frequency grid. Working in log-frequency space ensures that the kernel bandwidth is perceptually uniform across the frequency range. After smoothing, every point on the frequency grid receives a value, providing a complete and continuous profile over the full extent of the cochlear array.

NH Reference Bell. The same KDE procedure was applied to NH listeners, with their Gammatone-spaced stimulus values remapped to the nearest frequency grid points. A group-level NH reference bell $\text{bell}_{\text{NH}}(f, v)$ was computed by averaging individual NH bells across all NH participants, yielding the expected distribution of confident responses for a listener with normal cochlear place coding.

Delta Bell: Displacement Relative to the NH Reference. To quantify the shift between each CI user's confidence profile and the NH reference, a delta bell was computed

Preprint, not peer-reviewed

for each participant, vowel, and formant dimension as the signed difference between the participant's normalised bell and the group-level NH reference bell:

$$\Delta\text{bell}(f, v) = \text{bell}_{\text{CI}}(f, v) - \text{bell}_{\text{NH}}(f, v) \tag{3.3}$$

where both bells have been normalised to unit area prior to subtraction, ensuring that differences reflect shape and location rather than overall response count. Positive values at a given frequency f indicate that the CI user assigns greater relative confidence at that cochlear location than the NH reference, and negative values indicate reduced relative confidence. The delta bell therefore captures the direction and magnitude of the perceptual displacement induced by the CI signal relative to normal cochlear place coding, providing the dependent variable for Study Question 2.

3.2.6 Data Extraction and Statistical Analysis

All statistical analyses were conducted in MATLAB R2025a at a significance threshold of $\alpha = .05$. Two families of analyses were conducted, corresponding to the two study questions introduced in Section 3.1.

Study Question 1 - How do Vowel Representations Differ between Groups?

Three complementary analyses were applied to the electrode confidence bell to characterise group differences in vowel representation, each addressing a distinct aspect of the distributional difference between CI users and NH listeners.

Wasserstein Distance. To assess how far CI users' electrode confidence distributions deviate from those of NH listeners, the Wasserstein distance was computed for each CI participant, vowel, and formant dimension. The Wasserstein distance quantifies the minimum work required to transform one probability distribution into another, and provides a single interpretable scalar - here expressed in octaves - representing how far a CI user bell distribution is displaced from the NH reference along the frequency axis, irrespective of direction. For the NH group, a leave-one-out variant was used: each NH participant's bell was compared to the mean bell of the remaining NH participants, providing a within-group baseline. This procedure ensures that the NH distances reflect genuine inter-individual variability rather than comparison against a distribution that includes the participant's own data. A linear mixed model (LMM) was then fitted to the resulting distances:

$$W_{ivj} = \beta_0 + \beta_{\text{group}} \cdot \text{Group}_j + \beta_{\text{vowel}} \cdot \text{Vowel}_i + \beta_{\text{formant}} \cdot \text{Formant}_v + u_j + \varepsilon_{ivj} \tag{3.4}$$

where W_{ivj} is the Wasserstein distance for participant j, vowel i, and formant v; $u_j \sim \mathcal{N}(0, \sigma_u^2)$ is the participant-level random intercept; and ε_{ivj} is the residual. Group effects were evaluated via Type III ANOVA on the fitted model, and pairwise contrasts between NH, Post-LCI, and Pre-ECI were conducted via model-based contrasts (`coefTest`).

Preprint, not peer-reviewed

Formant Shift. Whereas the Wasserstein distance captures the overall magnitude of distributional displacement, it does not reveal its direction. To characterise where along the frequency axis CI users' perceptual representations are shifted relative to NH, the weighted centroid of each participant bell was computed as:

$$\hat{f}(v) = 10^{\dfrac{\sum_f \log_{10}(f) \cdot \mathrm{bell}(f,v) \cdot \Delta(\log f)}{\sum_f \mathrm{bell}(f,v) \cdot \Delta(\log f)}} \tag{3.5}$$

where the summation is over all frequency grid points f and $\Delta(\log f)$ is the log-spaced integration weight. The formant shift for each participant, vowel, and formant dimension was then defined as:

$$\Delta_{\text{shift}} = \log_2 \left(\frac{\hat{f}_{\text{NH}}(v)}{\hat{f}_{\text{CI}}(v)} \right) \tag{3.6}$$

where $\hat{f}_{\text{NH}}(v)$ is the NH reference centroid for vowel v (computed as the mean centroid across NH participants using the same leave-one-out procedure described above) and $\hat{f}_{\text{CI}}(v)$ is the CI participant's centroid. Positive values indicate that the CI participant's perceptual representation is centred at a more apical electrode location than NH (*i.e.*, the CI user perceives the formant at a lower frequency than expected); negative values indicate a more basal representation. Group differences in formant shift were assessed via Wilcoxon rank-sum tests with Holm-Bonferroni correction for multiple comparisons, applied to per-participant mean shifts (one value per participant, averaged across vowels and formants). Pairwise comparisons were conducted between all three groups (NH, Post-LCI, and Pre-ECI).

F1/F2 Reliance Ratio. To quantify the asymmetry of perceptual confidence between the F1 and F2 electrode regions, a reliance ratio R was computed for each participant and each vowel. For a given vowel, the reliance ratio is defined as:

$$R(v) = \frac{\mathrm{peak}\big(\mathrm{bell}_{F2}(f,v)\big)}{\mathrm{peak}\big(\mathrm{bell}_{F1}(f,v)\big) + \mathrm{peak}\big(\mathrm{bell}_{F2}(f,v)\big)} \tag{3.7}$$

where $\mathrm{bell}_{F1}(f,v)$ and $\mathrm{bell}_{F2}(f,v)$ are the confidence bells built from trials on which the F1 and F2 electrodes, respectively, were stimulated for vowel v, and $\mathrm{peak}(\cdot)$ denotes the maximum value of the bell over the frequency grid. This formulation uses the peak height of each bell rather than its integral, because both bells are normalised to unit integral by construction; the peak height therefore captures the concentration - and hence the reliability - of each formant's representation. Critically, this measure requires no frequency boundary to separate the F1 and F2 regions: the F1/F2 assignment follows directly from which electrode was stimulated in each trial, making the ratio valid across all vowels regardless of their acoustic formant frequencies. A value of $R > 0.5$ indicates that a participant places greater confidence in their F2 electrode responses than in their F1 electrode responses (F2 dominance); $R < 0.5$ indicates F1 dominance. The overall reliance ratio per participant was obtained by averaging $R(v)$ across vowels. Group differences in R were assessed via Wilcoxon rank-sum tests with Holm-Bonferroni correction.

Preprint - not peer-reviewed

Study Question 2 - Do Frequency-to-Place Factors Predict the Displacement of Electrode Confidence Relative to Normal-Hearing Listeners?

To test whether FTPM, CS, and NR predict the direction and magnitude of the shift between each CI user's confidence profile and the NH reference, electrode-level LMMs were fitted with the delta bell height as the outcome:

$$\Delta\text{bell}_{iejv} = \beta_0 + \beta_{\text{factor}} \cdot \text{Factor}_e + \beta_{\text{vowel}} \cdot \text{Vowel}_i + u_j + u_e + \varepsilon_{iejv} \tag{3.8}$$

where Δbell_{iejv} is the signed difference between participant j's normalised bell and the NH reference bell at electrode e, for vowel v and formant dimension i (Equation 3.3); Factor_e denotes one of FTPM_e, CS_e, or NR_e at electrode e; u_j is the participant random intercept; and u_e is the electrode random intercept, which accounts for the spatial autocorrelation introduced by KDE smoothing (Equation 3.2). A positive coefficient $\hat{\beta}_{\text{factor}}$ indicates that electrodes with higher factor values are associated with a positive displacement relative to the NH reference (greater relative confidence than NH listeners at those cochlear locations); a negative coefficient indicates displacement in the opposite direction. Because the three factors are available for different subsets of participants (see Section 3.2.4), separate univariate models were fitted for each factor to avoid listwise deletion. F1 and F2 were modelled separately. To test whether these relationships differed between Pre-ECI and Post-LCI groups, a Group × Factor interaction term was added as an extension of each univariate model, evaluated via likelihood ratio test. It should be noted that the small and unequal subsamples available for each factor (see Table 3.1) severely limit the power of this interaction test; the results should therefore be interpreted as exploratory. As an exploratory complement, the absolute cochlear position of each electrode - expressed as $\log_2(\text{CF}_{\text{place}})$, where CF_{place} is the tonotopic characteristic frequency of the stimulated cochlear region - was substituted for FTPM in an otherwise identical LMM. To determine whether absolute cochlear location or relative mismatch is the stronger predictor of confidence displacement, the two models were compared using the Akaike information criterion (AIC): a difference of more than two AIC units in favour of the CF_{place} model was taken as evidence that absolute cochlear location explains variance in the displacement beyond what is captured by relative mismatch alone. This comparison was performed separately for the F1 and F2 frequency regions.

3.3 Results

3.3.1 Overview

Identification and Confidence Distributions

The distribution of vowel responses and the distribution of confidence ratings across the three listener groups are displayed in Figures 3.6 and 3.7, respectively.

For NH listeners, responses were distributed relatively evenly across four of the five vowel categories ([ɑː]: 20.7%, [ɜː]: 23.1%, [uː]: 19.8%, [ɔː]: 21.0%), with the exception of [iː], which was selected on only 9.0% of trials. A broadly similar pattern was observed for Pre-ECI ([ɑː]: 28.8%, [ɜː]: 22.7%, [uː]: 20.9%, [ɔː]: 16.2%, [iː]: 11.5%) and Post-LCI ([ɑː]: 31.5%, [ɜː]: 17.9%, [uː]: 17.2%, [ɔː]: 23.8%, [iː]: 9.3%). No pairwise group differences reached significance for any vowel category (Kruskal-Wallis tests, all $p > .10$).

Preprint – Not peer reviewed

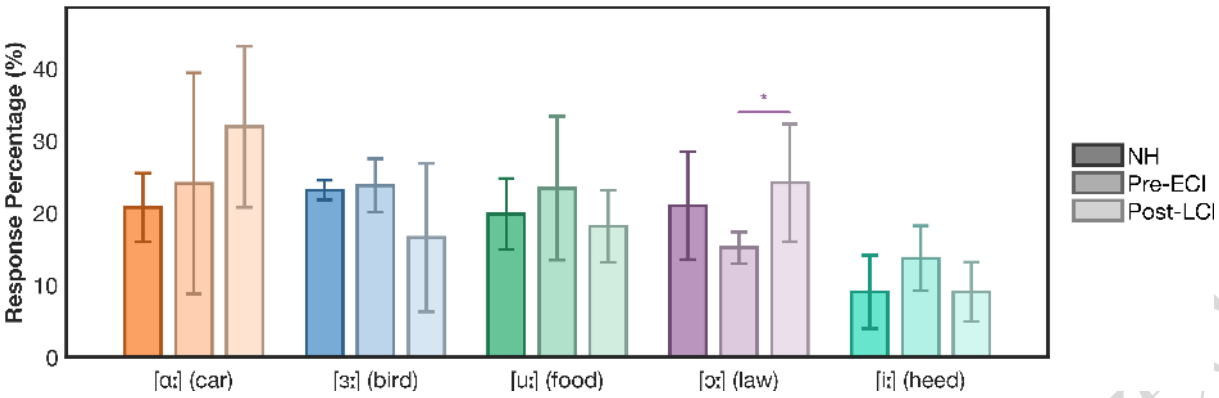


Figure 3.6: **Response distribution across the five vowel categories.** Percentage of trials on which each vowel was selected, shown separately for NH listeners ($n = 10$), Post-LCI ($n = 14$), and Pre-ECI ($n = 10$). Error bars indicate ±1 SD across participants.

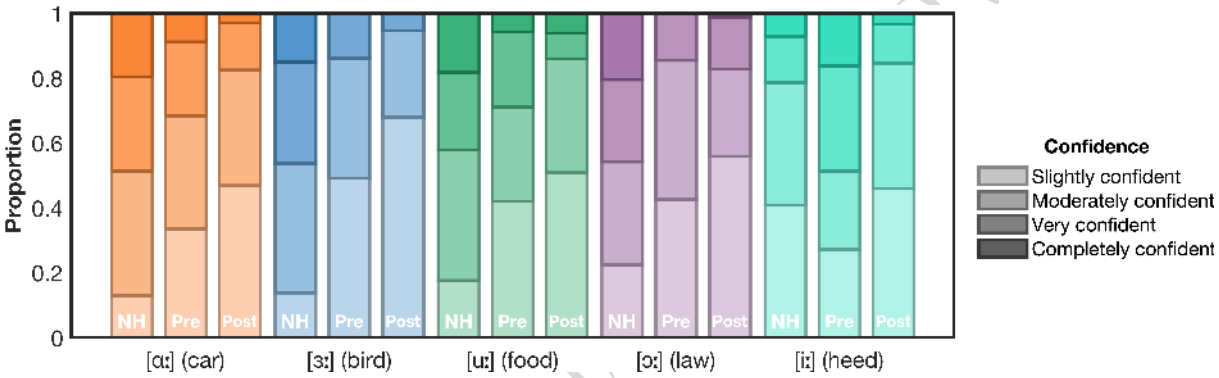


Figure 3.7: **Confidence rating distributions across the five vowel categories.** Proportion of responses in each confidence category, shown separately for the three groups. Both CI groups showed overall lower confidence than NH listeners, with a larger proportion of responses in the lowest confidence category across most vowels.

Visual inspection of confidence ratings confirmed that the distribution was broadly consistent across vowels within each group, with the exception of [iː]. For NH listeners, [iː] responses were characterised by a marked drop in confidence: 40.8% of [iː] responses fell in the lowest confidence category, compared with 12.8–22.3% for the other vowels. Both CI groups showed overall lower confidence than NH listeners across most vowels (Pre-ECI: 25–53%; Post-LCI: 45–65% in the lowest category).

Perceptual Maps and Exclusion of [iː]

Figure 3.8 presents an example of the Bayesian confidence maps for a representative CI participant alongside the mean NH reference, for all five vowel categories. For most vowels, differences between the CI user and the NH groups were subtle, with comparable F1 and F2 ranges and broadly overlapping confidence regions.

The exception was [iː]. Three converging observations motivated its exclusion from subsequent analyses. First, [iː] was selected on fewer than 10% of trials across all three groups, substantially less than the other four vowels (Section 3.3.1). Second, when [iː]

Preprint – Not peer reviewed

was selected, confidence ratings were markedly lower than for the other vowels in the NH group, with over 40% of responses falling in the lowest confidence category. Third, the perceptual maps revealed a systematic displacement of the F1 representation in CI users: for the NH group, the [iː] confidence map showed a peak at high F2 values (approximately 2000–4000 Hz) and low F1 values (below approximately 500 Hz), consistent with its expected formant structure; for the CI user, the F1 of [iː] responses was displaced upward to approximately 2500 Hz - close to the F2 range - while F2 remained comparable to the NH group. This pattern was observed consistently across CI participants, suggesting that CI users may have perceived the two formants as a single auditory object, or interpreted the stimulus as containing F2 and F3 rather than F1 and F2. The heatmap figure below includes [iː] for completeness, but all quantitative analyses reported in the remainder of this section concern the four retained vowel categories: [ɑː], [ɜː], [ɔː], and [uː].

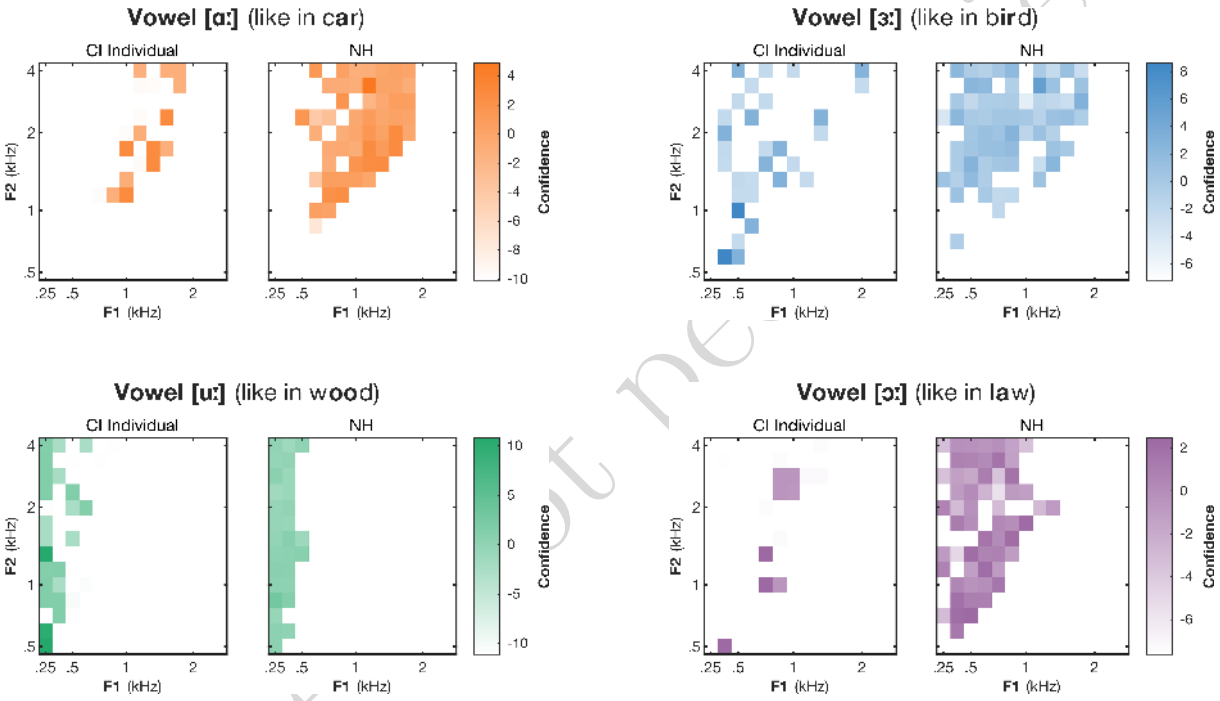


Figure 3.8: **Perceptual maps for a CI participant.** Each panel displays the median confidence surface across the F1×F2 formant space for one vowel category.

3.3.2 Study Question 1 - How do Vowel Representations Differ between Groups?

The electrode confidence distributions (bells) for each group and vowel are displayed in Figure 3.9. Three complementary analyses were applied to characterise group differences: the Wasserstein distance (overall magnitude of displacement from NH), the formant centroid shift (direction of displacement), and the F1/F2 reliance ratio (asymmetry of confidence across the frequency axis).

Preprint, not peer reviewed

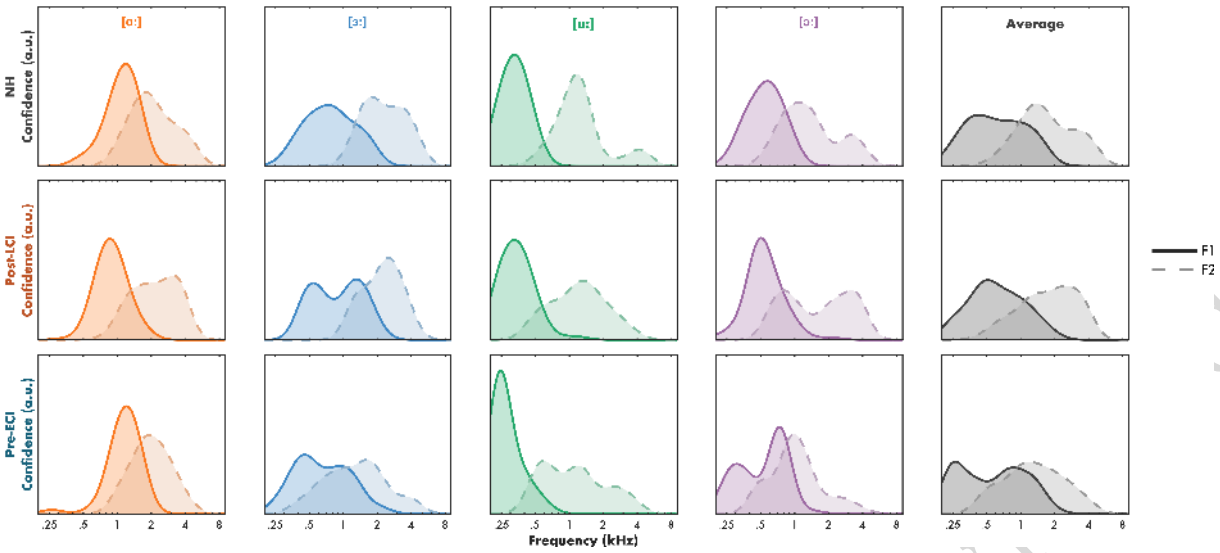


Figure 3.9: **Electrode confidence distributions (bells) for NH, Post-LCI, and Pre-ECI listeners.** Each panel shows the confidence-weighted kernel density estimate over the frequency axis for one group (rows) and one vowel (columns), with F1 (solid) and F2 (dashed) plotted separately. The rightmost column shows the average bell across vowels. The x-axis is in log-frequency (kHz).

Wasserstein distance from the NH reference

Figure 3.10 shows the Wasserstein distances and Table 3.2 summarises the LMM fixed effects. Note that the NH values shown in Figure 3.10 are not zero because they reflect leave-one-out distances: each NH participant's bell is compared to the mean of the remaining NH participants, providing a within-group variability baseline rather than a fixed reference point.

Preprint - not peer-reviewed

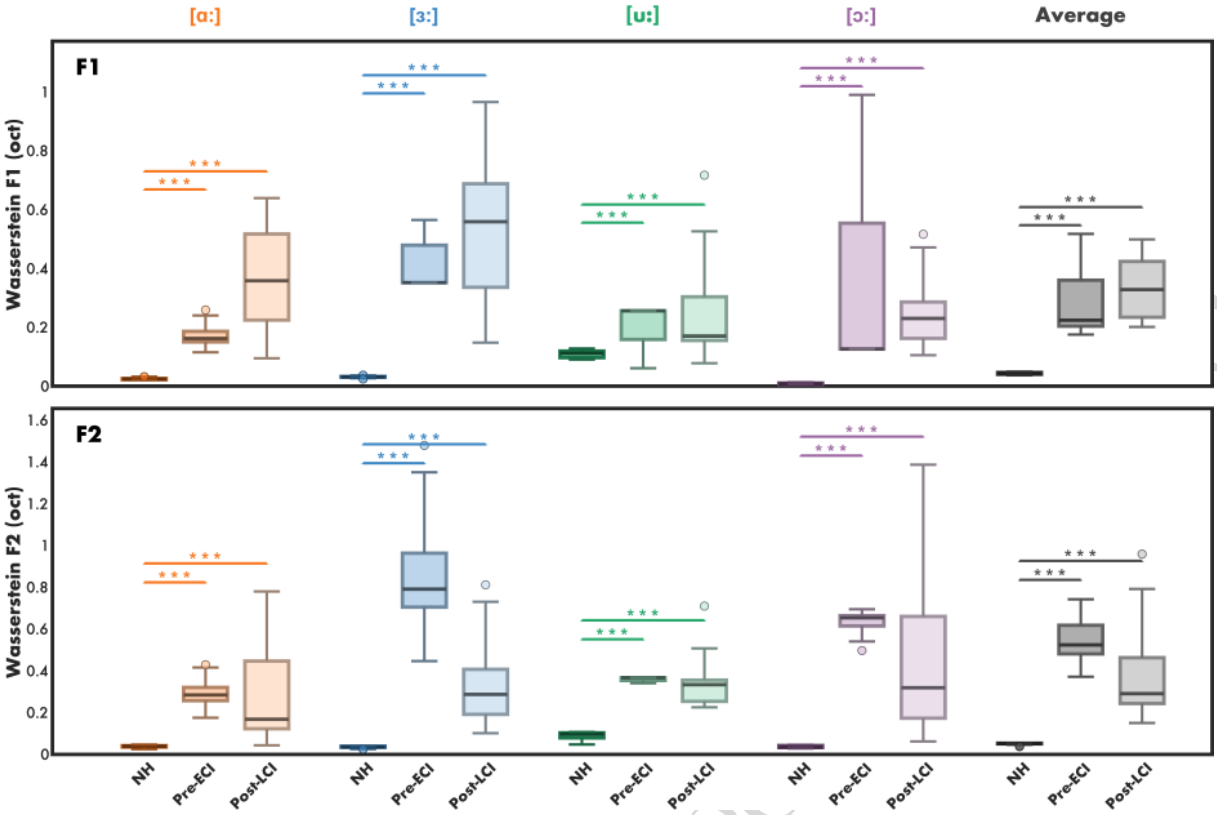


Figure 3.10: **Wasserstein distances from the NH reference bell, per vowel and formant.** Each box shows the distribution across participants for NH (leave-one-out), Post-LCI, and Pre-ECI. Significance bars indicate pairwise contrasts from the LMM ($^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$, ns: not significant).

Table 3.2: **LMM fixed effects: Wasserstein distance from the NH reference.** Model: Wasserstein $\sim$ Group + Vowel + Formant + (1|Participant). Reference levels: Group = NH, Vowel = [ɑ:], Formant = F1. $n = 209$ observations from 27 participants. The leave-one-out procedure for NH participants is described in Section 3.2.6.

Term	$\hat{\beta}$	SE	t	p	
Intercept	−0.036	0.033	−1.09	.276	
Group: Post-LCI	+0.306	0.029	+10.46	$< .001$	***
Group: Pre-ECI	+0.371	0.037	+10.02	$< .001$	***
Vowel: [ɜ:]	+0.120	0.038	+3.16	.002	**
Vowel: [ɔ:]	+0.059	0.037	+1.62	.108	
Vowel: [u:]	+0.030	0.037	+0.81	.418	
Formant: F2	+0.061	0.026	+2.31	.022	*
ANOVA (Type III):					
Group	$F(2, 201) = 76.00$			$< .001$	***
Vowel	$F(3, 201) = 3.59$			.015	*
Formant	$F(1, 201) = 5.32$			.022	*

Group was a highly significant predictor of Wasserstein distance ($F(2, 201) = 76.00$, $p < .001$). Both CI groups were substantially displaced from the NH leave-one-out base-

Preprint not peer reviewed

line: Post-LCI ($\hat{\beta} = +0.306$ oct, $p < .001$) and Pre-ECI ($\hat{\beta} = +0.371$ oct, $p < .001$). Crucially, Post-LCI and Pre-ECI did not differ from each other ($F(1, 201) = 2.95$, $p = .088$), indicating that the two groups are displaced from NH to a statistically equivalent degree regardless of their implantation history. Vowel was also significant ($F(3, 201) = 3.59$, $p = .015$), with [ɜː] showing the greatest displacement ($\hat{\beta} = +0.120$ oct, $p = .002$). Formant was significant ($F(1, 201) = 5.32$, $p = .022$), with F2 displaced further from NH than F1 ($\hat{\beta} = +0.061$ oct).

Formant Centroid Shift

Figure 3.11 displays the frequency centroid shift in octaves for each group, vowel, and formant. Positive values indicate that a participant's confidence distribution is centred at a more apical cochlear location than the NH reference (*i.e.*, the CI user perceives the formant at a lower frequency than expected).

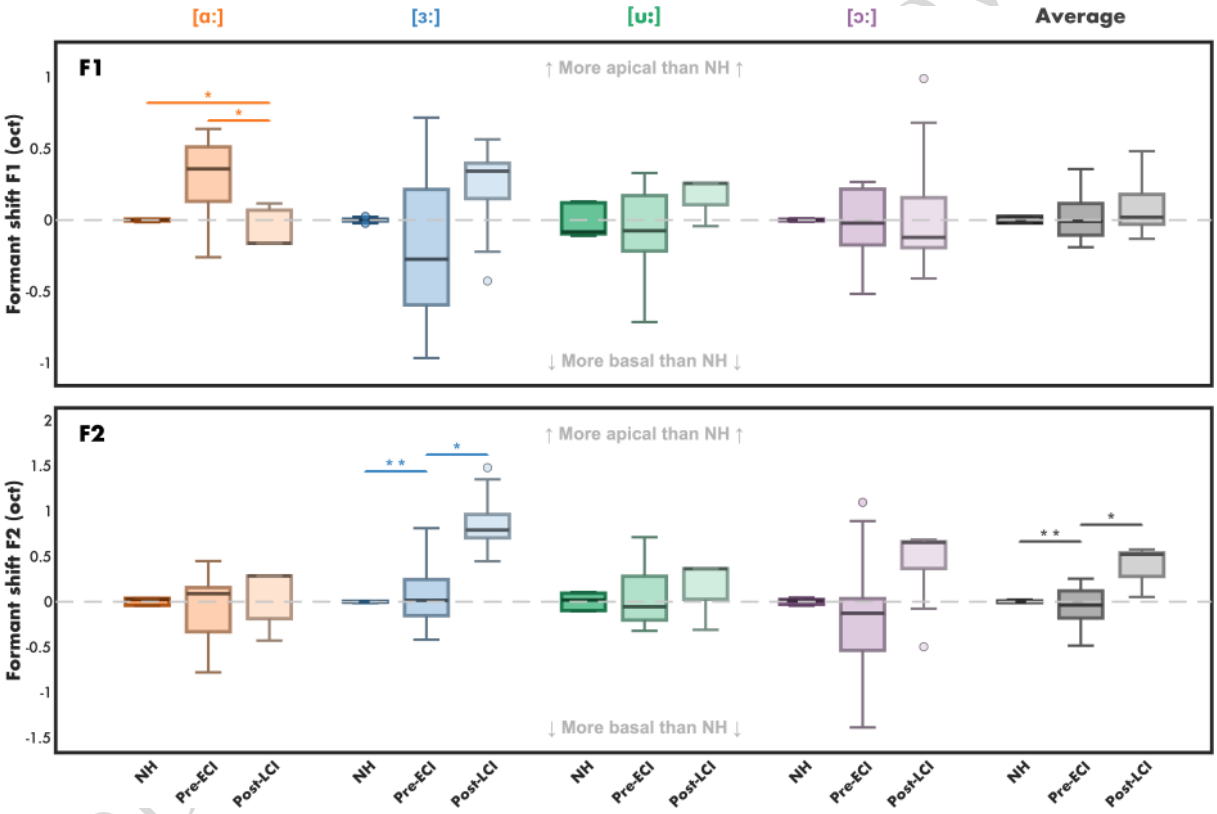


Figure 3.11: **Formant centroid shift relative to the NH reference, per vowel.** Shift = $\log_2(\hat{f}_{\text{NH}}/\hat{f}_{\text{CI}})$ in octaves; positive values indicate the CI user's distribution is centred at a lower frequency than NH (more apical). For NH, leave-one-out centroids are used. Significance bars indicate pairwise Wilcoxon rank-sum tests with Holm-Bonferroni correction.

For F1, both CI groups showed a slight tendency towards apical shift relative to NH, consistent with the known apical insertion bias of CI electrode arrays. For F2, a significant difference was found between the two CI groups and between the NH listener group and the Pre-ECI group, which exhibited a preference towards lower F2 frequencies. However,

Pre-reviewed

this observation at the average level should be interpreted with caution because it was largely driven by a single vowel, namely [ɜː].

F1/F2 Reliance Ratio

Figure 3.12 shows the F1/F2 reliance ratio for each group and vowel. A value of $R > 0.5$ indicates that a participant placed greater confidence in their F2 electrode responses than in their F1 electrode responses.

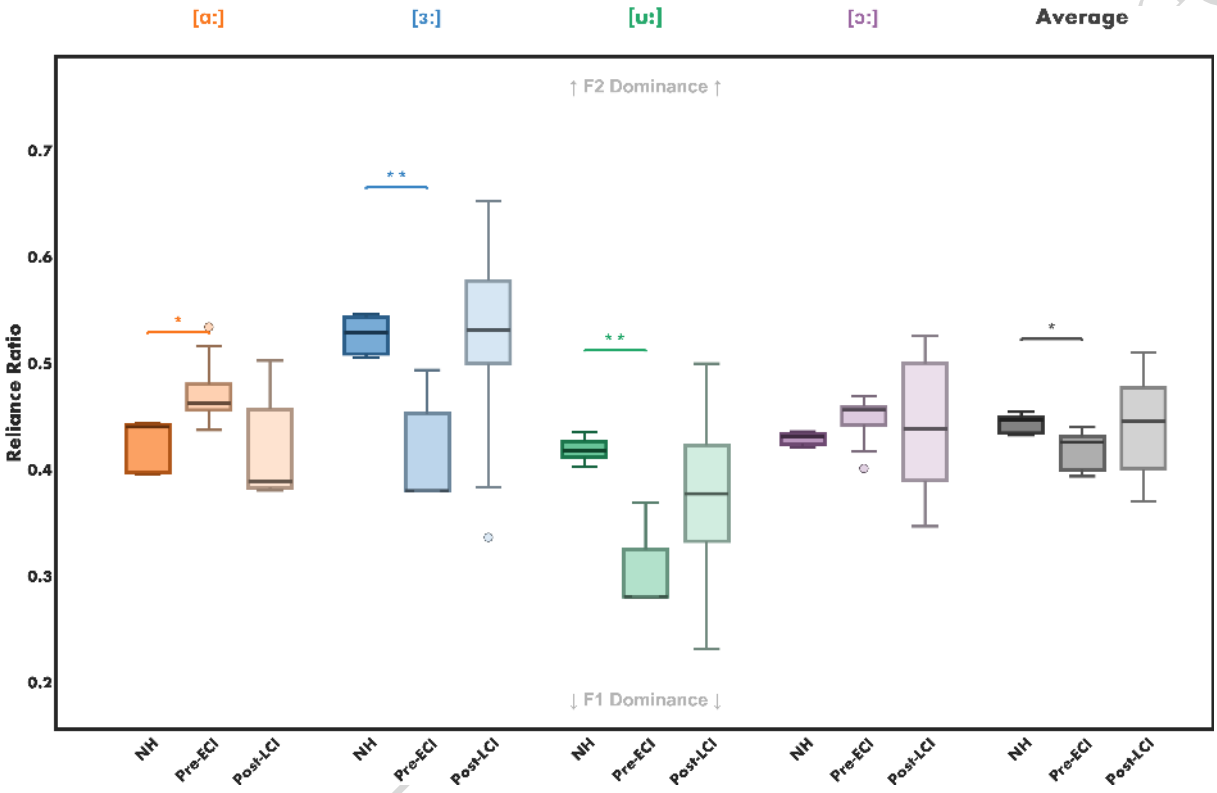


Figure 3.12: **F1/F2 reliance ratio per group and vowel.** $R = \text{peak}(\text{bell}_{F2})/(\text{peak}(\text{bell}_{F1}) + \text{peak}(\text{bell}_{F2}))$. Values above 0.5 indicate F2 dominance; values below 0.5 indicate F1 dominance. Significance bars indicate pairwise Wilcoxon rank-sum tests with Holm-Bonferroni correction ($^{*}p < .05$, $^{**}p < .01$, ns: not significant).

Descriptively, NH listeners showed moderate F1 dominance on average ($R = 0.443 \pm 0.008$), consistent with the fact that most English vowels are differentiated primarily by F1 (Peterson and Barney, 1952; Hillenbrand et al., 1995). Post-LCI listeners showed a similar mean reliance ($R = 0.439 \pm 0.046$), whereas Pre-ECI listeners showed a lower ratio ($R = 0.418 \pm 0.019$), suggesting a stronger tendency towards F1 dominance in Pre-ECI users. The within-group standard deviations were substantially larger for the CI groups than for NH, reflecting the considerable inter-individual variability in formant reliance among CI users.

Pairwise comparisons revealed that Pre-ECI differed significantly from NH for [ɑː] (adjusted $p = .037$), [ɜː] (adjusted $p = .001$), [uː] (adjusted $p = .004$), and at the mean level (adjusted $p = .026$), but not for [ɔː] ($p = .203$). Post-LCI did not differ significantly from NH for any vowel or at the mean level (all adjusted $p \geq .149$), nor did Post-LCI

Preprint - Submitted

differ from Pre-ECI (all adjusted $p \geq .059$).

These results indicate that Pre-ECI users show a distinct pattern of spectral reliance: they weight F1 electrode responses more heavily than both NH listeners and Post-LCI users, particularly for vowels whose identity depends on high F2 frequency ([ɜː], [uː]). This asymmetry was not present in Post-LCI users, suggesting that prior auditory experience with normal hearing may have enabled Post-LCI users to maintain a more balanced reliance across the frequency axis.

3.3.3 Study Question 2 - Do Frequency-to-Place Factors Predict the Displacement of Electrode Confidence Relative to NH Listeners?

Figure 3.13 displays the delta confidence as a function of each factor, pooled across all CI participants with factor data. At each frequency location, the delta confidence reflects the signed deviation of a CI user's perceptual confidence from the NH reference bell at the same frequency position. Univariate LMMs were fitted separately for each factor and formant, first on the pooled CI sample and then within each group. Results are summarised in Tables 3.3, 3.4, and 3.5.

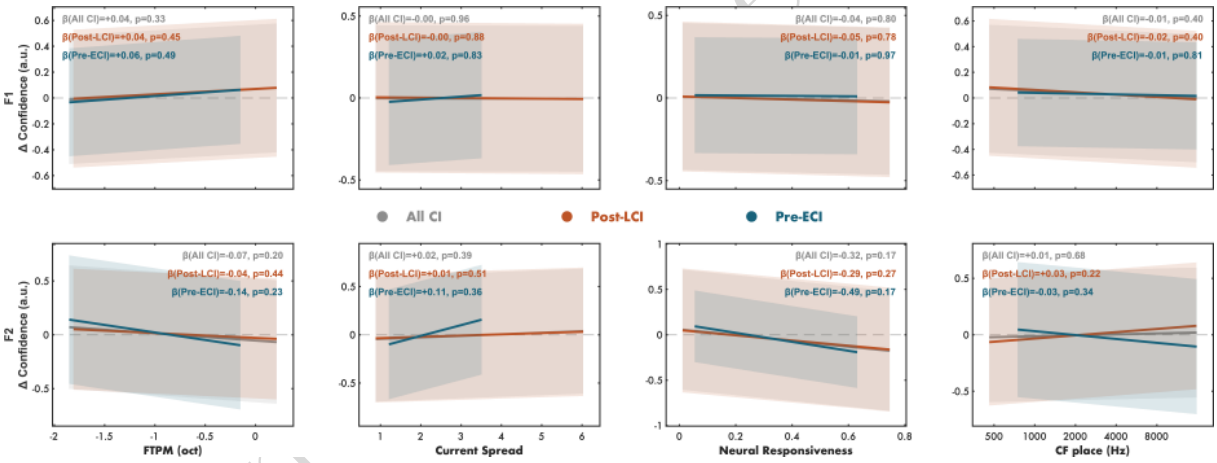


Figure 3.13: **Local confidence displacement ($\Delta = \text{bell}_{\text{CI}} - \text{bell}_{\text{NH}}$) as a function of FTPM, CS, NR, and CF_{place}.** Each panel shows the difference between the CI and NH-reference bell height at matched cochlear positions for F1 (top row) and F2 (bottom row). Grey: all CI participants; orange: Post-LCI; blue: Pre-ECI. A value of zero indicates no deviation from NH; negative values indicate that CI users place less confidence at that electrode than NH listeners. Shaded bands represent ±1 residual SD around the regression line.

Pooled Sample

None of the electrode-level factors significantly predicted the delta confidence for either F1 or F2 in the pooled CI sample (all $p \geq .191$; Table 3.3). This null result held across FTPM, CS, NR, and CF_{place}, for both formants.

Preprint, peer-reviewed

Table 3.3: **LMM fixed effects: delta confidence ($\Delta = \mathbf{bell_{CI} - bell_{NH}}$) - pooled CI sample.** Model: $\Delta \sim$ Factor + Vowel + (1|Participant) + (1|Electrode).

Formant	Factor	n	$\hat{\beta}$	SE	t	p
F1	FTPM	514	+0.040	0.045	+0.88	.380
F1	CS	388	−0.000	0.013	−0.00	.997
F1	NR	300	−0.040	0.151	−0.26	.793
F1	$\log_2(\mathrm{CF}_{\mathrm{place}})$	514	−0.011	0.019	−0.57	.568
F2	FTPM	514	−0.066	0.052	−1.27	.204
F2	CS	388	+0.016	0.019	+0.87	.385
F2	NR	300	−0.303	0.231	−1.31	.191
F2	$\log_2(\mathrm{CF}_{\mathrm{place}})$	514	+0.008	0.020	+0.41	.684

Post-LCI Group

No factor significantly predicted delta confidence in the Post-LCI group for either F1 or F2 (all $p \geq .289$; Table 3.4). Effect sizes and directions were broadly consistent with the pooled pattern, with the largest (non-significant) effect being a negative NR slope for F2 ($\hat{\beta} = -0.281$, $p = .289$), directionally consistent with the pooled result.

Table 3.4: **LMM fixed effects: delta confidence - Post-LCI group.** Model: $\Delta \sim$ Factor + Vowel + (1|Participant) + (1|Electrode).

Formant	Factor	n	$\hat{\beta}$	SE	t	p
F1	FTPM	346	+0.048	0.055	+0.87	.386
F1	CS	308	−0.001	0.014	−0.06	.953
F1	NR	272	−0.044	0.168	−0.26	.796
F1	$\log_2(\mathrm{CF}_{\mathrm{place}})$	346	−0.014	0.026	−0.55	.581
F2	FTPM	346	−0.045	0.058	−0.78	.438
F2	CS	308	+0.015	0.021	+0.71	.477
F2	NR	272	−0.281	0.264	−1.06	.289
F2	$\log_2(\mathrm{CF}_{\mathrm{place}})$	346	+0.028	0.025	+1.12	.262

Pre-ECI Group

No factor significantly predicted delta confidence in the Pre-ECI group (all $p \geq .107$; Table 3.5). The largest effect was a positive FTPM slope for F1 ($\hat{\beta} = +0.156$, $SE = 0.096$, $p = .107$), which did not reach significance and was in the opposite direction to the prediction that greater mismatch would reduce confidence.

Preprint - not peer-reviewed

Table 3.5: **LMM fixed effects: delta confidence - Pre-ECI group.** Model: $\Delta \sim$ Factor + Vowel + (1|Participant) + (1|Electrode).

Formant	Factor	n	$\hat{\beta}$	SE	t	p
F1	FTPM	168	+0.156	0.096	+1.62	.107
F1	CS	80	+0.018	0.080	+0.22	.823
F1	NR	28	−0.012	0.290	−0.04	.966
F1	$\log_2(\mathrm{CF}_{\mathrm{place}})$	168	−0.006	0.025	−0.24	.811
F2	FTPM	168	−0.140	0.116	−1.20	.231
F2	CS	80	+0.111	0.127	+0.88	.383
F2	NR	28	−0.495	0.541	−0.92	.370
F2	$\log_2(\mathrm{CF}_{\mathrm{place}})$	168	−0.034	0.036	−0.96	.338

3.4 Discussion

This study examined how experienced CI users represent vowels in the F1×F2 frequency space and whether frequency-to-place factors could account for deviations from the NH reference. Two groups of CI users were compared according to their hearing history and implantation timing: Pre-ECI and Post-LCI.

Three principal findings emerge. First, both CI groups showed electrode confidence distributions significantly displaced from the NH reference to a statistically equivalent degree. In terms of shift, F2 was shifted towards lower frequencies for Post-LCI users relative to NH and Pre-ECI; however, this shift was relatively moderate and not generalised across all vowels. Second, the two CI groups differed in the relative weighting they assigned to F1 and F2: Post-LCI users maintained a balanced F1/F2 reliance ratio comparable to NH, whilst Pre-ECI users showed F1 dominance on average, suggesting that the two groups differ in their reliance on spectral cues from different cochlear regions. Third, none of the frequency-to-place factors predicted a local confidence deficit relative to NH - a null result that mirrors the absence of systematic vowel category shifts reported in experienced CI users. These results may reflect complete group-level adaptation to the CI signal. Each finding is discussed in turn.

3.4.1 Experienced CI Users Differ Modestly from NH Listeners in Vowel Representation

A natural initial expectation, shared by Harnsberger et al. (2001), is that Post-LCI users - who underwent a remapping process since implantation - should show residual displacement from NH, whilst Pre-ECI users - who built their representations entirely around the CI input - should show a more NH-like pattern.

The present data do not diverge substantially from these expectations: even though the Wasserstein analysis reported a significant displacement between NH listeners and CI user responses, both the shift and reliance analyses clearly mitigate that displacement, with differences restricted to a subset of vowels and not systematic across the full set.

These modest differences are consistent with previous data indicating that the per-

Preprint - not peer-reviewed

ception of vowels in the frequency space does not differ substantially between experienced CI users and NH listeners. Harnsberger et al. (2001) examined vowel perception in the F1×F2 space of CI users using a method-of-adjustment paradigm and found that seven of eight experienced Post-LCI users showed no systematic shift in vowel category centres relative to NH, concluding that most users had successfully adapted to their device's basalward spectral shift. The small differences between Harnsberger et al. (2001) and the present data are likely attributable to methodological differences rather than genuine perceptual divergence. In our task, participant responses are constrained to the electrode mapping: even though the NH F1×F2 space is individually rescaled to match each CI user's electrode layout, the discrete nature of electrode positions limits the available response options and may partly account for the residual differences observed between NH listeners and CI users.

3.4.2 Pre-ECI Users Show Greater Reliance on Apical Electrodes than Post-LCI Users

Despite equivalent overall displacement from NH, the two CI groups differed in a clinically important respect. Whereas Post-LCI users maintained a balanced F1/F2 reliance ratio comparable to NH ($R_{\text{LCI}} = 0.439 \pm 0.046$ vs $R_{\text{NH}} = 0.443 \pm 0.008$), Pre-ECI users showed F1 dominance for [ɜː], [uː], and on average (all adjusted $p \leq .037$). Two interpretations of this finding should be considered. The first is that Pre-ECI users, having developed their auditory system around electrical input from birth, learned to extract information from apical electrodes more effectively than Post-LCI users. The second - and arguably more parsimonious - interpretation is that greater F1 reliance reflects degraded F2 encoding rather than superior F1 processing: if the spectral quality of F2 cues is poorer for Pre-ECI users (for example, due to reduced neural survival or broader current spread in the basal cochlear regions encoding F2), compensatory reliance on the better-preserved F1 channel would produce the same pattern without implying any advantage in apical processing. Supporting this second interpretation, NH listeners - who have access to both formants at full fidelity - showed a balanced reliance ratio ($R = 0.443$), indicating that when F2 cues are reliably encoded, listeners do not preferentially weight F1. The present data cannot distinguish between these two accounts; the PECAP data available for the small subset of Pre-ECI participants with both CS and NR estimates did not reveal systematic differences in basal neural health between groups, though the sample was too small to draw firm conclusions. This finding illustrates one advantage of the electrode-level design adopted here: because each response is associated with a specific pair of electrodes, the task can reveal asymmetries in spectral reliance that would be invisible in conventional identification-only paradigms.

The difficulty of apical electrode coding has been well documented in the CI literature, and the relevant observations have been made predominantly in Post-LCI populations. At the electrode-array level, Gifford et al. (2022) showed that deeper insertion of a lateral wall array benefits word recognition scores (CNC words in quiet) up to approximately 600° of AID, with performance reaching a plateau thereafter, suggesting a limit on the perceptual utility of the most apical contacts. This plateau is consistent with physiological constraints specific to the apical spiral ganglion: unlike the basal and middle turns, the apical region contains ganglion cells of closely grouped characteristic frequencies, meaning that electrical stimulation from a single electrode activates neural populations that are

Preprint - not peer-reviewed

not well separated in frequency, reducing channel specificity (Faulkner, 2011). One study even reports that deactivating the most apical electrode in the Post-LCI population can result in an improvement of speech performance (Berg et al., 2025b). However, not all apical stimulation is necessarily detrimental: Berg et al. (2025b) speculated that some CI listeners may be able to exploit temporal interactions between channels, but only in lower-frequency (apical) regions where neural activation patterns overlap more extensively. This raises the possibility that Pre-ECI users, having developed their auditory system around electrical input from birth, may have learned to exploit these temporal interactions in a way that Post-LCI users - whose auditory cortex was shaped by acoustic hearing - cannot.

The present finding that Pre-ECI users relied more heavily on F1 than Post-LCI users is broadly coherent with previous observations concerning spectral cue utilisation in Pre-ECI versus Post-LCI populations, though the direction of the group differences does not map onto the cited literature in a straightforward way. DiNino and Arenberg (2020) directly compared acoustic cue weighting for speech contrasts in Pre-ECI and Post-LCI adults, and found that Pre-ECI users better utilised fine-grained formant cues, with formant cue weights correlating with vowel recognition. Giezen et al. (2010) reported that Pre-ECI users adopted cue-weighting strategies resembling those of NH children for phoneme contrasts, whereas Post-LCI users deviated from NH patterns and showed reduced weighting of spectral structure (Moberly et al., 2014). In the present study, it was Pre-ECI users who differed from NH, not Post-LCI users. This apparent discrepancy likely reflects differences in the perceptual dimension being measured: the cited studies assessed categorical cue weighting for phoneme identification, whereas the present task measures confidence-weighted spectral reliance across the electrode array, which conflates perceptual strategy with the quality of the signal available at each electrode. Nagels et al. (2024) further showed that Pre-ECI children weight both F0 and vocal-tract-length cues for voice gender categorisation similarly to NH children, in contrast to Post-LCI adults who rely predominantly on F0 and fail to use vocal-tract-length cues - which are encoded largely through formant frequencies - consistent with the reduced exploitation of fine-grained spectral cues in Post-LCI users suggested by the literature above.

These converging results support a common interpretation: whether through early cortical plasticity, the absence of competing acoustic representations, degraded encoding of higher-frequency spectral cues, or a combination of these factors, Pre-ECI users appear to have developed auditory strategies that rely more heavily on spectral structure in the low-frequency apical region than Post-LCI users.

3.4.3 Frequency-to-Place Factors do not Explain the Small Displacement

A further finding of the current study is the absence of any relationship between frequency-to-place factors and the small displacement in the perception of vowel representation between CI users and NH listeners. When the NH reference bell was subtracted at each matched electrode position, none of FTPM, CS, NR, or CF_{place} predicted the sign or magnitude of the local confidence displacement in the pooled sample (all $p \geq .191$), and this null result was consistent within both the Post-LCI (all $p \geq .289$) and Pre-ECI (all $p \geq .107$) subgroups.

This null result could reflect two distinct possibilities. First, the task may not be sen-

Preprint – not peer-reviewed

sitive enough or not adapted to probe individual differences in frequency-to-place factors. The task was designed to go beyond simple identification by incorporating a confidence rating as a proxy for perceptual clarity; however, speech-related tasks are known to be poor proxies for frequency-to-place factors, particularly after the adaptation period. Second, the CI users tested here may be fully adapted to these factors - a possibility supported by the fact that the differences from NH listeners are quantitatively small and restricted to specific vowels and conditions. Both interpretations are plausible and the present data cannot definitively distinguish between them. Chapter 5 may help clarify this: by testing a CI user longitudinally during the adaptation process with the same vowel task, it will be possible to examine whether the task can track changes in the perceptual process and reveal trends related to frequency-to-place factors during the period when adaptation is still ongoing.

3.4.4 Clinical Implications and Future Perspectives

Despite the small displacement of CI users and the absence of a relationship between frequency-to-place factors and vowel displacement, a few results stand out and may inform the clinical management of CI users and suggest directions for targeted mapping strategies.

First, we observed that, on average, the F2 formant was perceived at a lower frequency by Post-LCI users relative to Pre-ECI users and NH listeners. This finding converges with the growing literature on anatomy-based fitting: Creff et al. (2024) showed that tonotopic fitting - aligning electrode allocation to individual cochlear geometry via post-operative imaging - significantly improved speech recognition in noise without degrading performance in quiet, and Dessard et al. (2024); Venail et al. (2015b) found that larger mismatches between default and anatomy-based maps were associated with poorer speech performance and slower adaptation. Recent work by Sagi and Svirsky (2025) further suggests that the appropriate tonotopic target may itself depend on stimulation level, as cochlear tonotopy shifts basally at suprathreshold levels typical of conversational speech, raising questions about which frequency-place function should guide anatomy-based fitting. Taken together, these results suggest that the downward F2 shift observed here reflects a genuine perceptual preference for lower-frequency F2 allocation that could, in principle, be partially corrected through anatomy-informed remapping. Whether such remapping would be beneficial in experienced users - for whom the current allocation has already been adapted to - remains an open question; Svirsky et al. (2005) cautioned that changing an adapted map can temporarily disrupt performance even if the new map is ultimately better matched to cochlear tonotopy, and more recent work has confirmed that the costs of map changes can persist for weeks before adaptation restores performance.

For Pre-ECI users, the F1 dominance finding argues for restraint rather than intervention. If apical F1 electrodes are already well integrated into the perceptual strategy of Pre-ECI users, clinical interventions that deactivate these electrodes or shift the allocation away from the F1 region - motivated by their high FTPM values - may disrupt a well-adapted strategy.

Preprint - not peer-reviewed

3.4.5 Limitations

Several limitations qualify the conclusions drawn here. The electrode-level factor analyses involved small subsamples (eight participants for FTPM and CF_{place}, six for CS and NR as underlying factors), severely limiting statistical power and precluding per-vowel or per-group breakdowns.

An important methodological limitation is that participants made their F1 and F2 judgements jointly. Consequently, attributing the response to one specific electrode is limited both by the paradigm itself and by the analyses performed, which may partly explain the absence of any effect of individual frequency-to-place factors considered independently for each formant. One potential solution would be to consider them jointly - for example, as a ratio between F1-electrode and F2-electrode factors - though any such analysis would remain constrained by the fact that the confidence rating integrates information from both formants simultaneously. An experimental design alternative is discussed in the Postface of this chapter.

3.5 Conclusion

Using a vowel identification and confidence rating task in the F1×F2 perceptual space, we characterised and compared the vowel representations of two groups of experienced CI users - Post-LCI and Pre-ECI - relative to NH listeners. Differences between the three groups were small overall, yet Post-LCI users showed a preference for lower F2 frequencies relative to the other two groups, and Pre-ECI users weighted F1 more heavily in their judgements. We examined whether these differences could be explained by three frequency-to-place factors at the electrode level, but none of them accounted for the displacement of CI users relative to their NH peers. This null result may reflect either successful long-term adaptation of CI users to these factors, or insufficient sensitivity of the present task to detect their residual influence.

Preprint - not peer-reviewed

Postface

The paradigm presented in this chapter aimed to characterise the optimal vowel representation of experienced CI users by presenting different F1×F2 electrode combinations and asking participants to identify the vowel perceived alongside a continuous confidence rating. The task was thought as a proxy for spectral clarity, which is suspected to be degraded even in experienced users. However, the small differences observed between both CI groups and the NH listeners - and the absence of any relationship between frequency-to-place factors and those displacements - point to two possible interpretations: either the CI users tested were fully adapted to their device's spectral distortions, or the task lacked sufficient sensitivity to detect their residual influence. As Chapter 5 will show, both explanations are likely to be true simultaneously.

Improving the Experimental Design: from Single Sounds to Paired Comparisons. The most fundamental limitation of the present implementation is the use of a single stimulus per trial. Presenting one sound at a time makes the task considerably more demanding for participants and, crucially, conflates certainty about F1 and F2 in a single confidence rating - making it impossible to attribute a response to one formant independently of the other. This is probably one reason why electrode-level frequency-to-place factors did not reach significance in the individual-level analyses: each trial yields a single confidence value for a joint F1×F2 combination, so the respective contributions of the two electrodes cannot be fully disentangled.

A natural solution would be to move to a paired-comparison format, as already used successfully in Chapter 4. Two sounds would be presented per trial - either varying both F1 and F2 simultaneously, or varying only one formant at a time - and participants would judge which sounds more like a unique target vowel. An example interface for this alternative design is shown in Figure 3.14. The single-vowel paired-comparison structure would be easier for participants, and the Bayesian posterior analysis applied in Chapter 4 could be directly adapted here to handle the full set of pairwise comparisons efficiently, even with an adaptive presentation paradigm that reduces the total number of trials required. Varying only one formant per trial would additionally allow independent F1 and F2 bells to be estimated without post hoc decomposition, directly resolving the primary analytical limitation of the current paradigm.

Preprint - not peer-reviewed

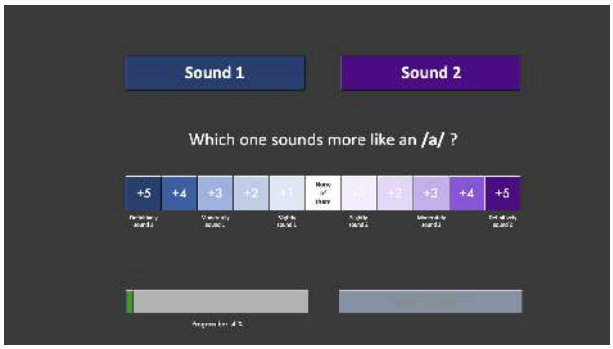


Figure 3.14: **Alternative interface for the vowel task.** In this design, two different F1×F2 combinations are presented per trial and the participant judges which sounds more like the target vowel. Although all pairwise combinations would in principle need to be tested, an adaptive presentation schedule would substantially reduce the number of trials required. The paired-comparison structure would be easier for participants and would allow Bayesian posterior estimation - as used in Chapter 4 - to be applied efficiently.

Beyond the current implementation and potential improvement of the design, the current paradigm opens several new scientific and clinical directions.

Personalised Remapping from Perceptual Data Alone. The formant centroid shift analysis provides, for each participant, a direct quantitative estimate of how far their F1 and F2 representations are displaced from the NH reference. This information could be used to derive a participant-specific frequency allocation adjustment without requiring post-operative CT imaging: if a user's F2 bell is centred 0.10 oct below NH, that directly quantifies the allocation shift needed to bring their perceptual representation into register. This approach would translate the research paradigm into a clinical tool accessible to any centre, bypassing the need for radiological data. It could be implemented during the adaptation procedure to target directly the best mapping for the participant from Day 1.

Cross-Population and Cross-Modality Extensions. We can imagine providing the test beyond the populations studied in the present study, for example with bimodal CI users, HA users, or people with specific hearing disorders.

Applied to electric-acoustic stimulation users, the current paradigm could identify the perceptual crossover position between acoustic and electric reliance - the cochlear location at which the listener transitions from acoustic low-frequency hearing to electrode-based encoding. Applied to HA users with progressive high-frequency loss, it could characterise how the effective perceptual bandwidth narrows as F2 information degrades, providing a continuum model that bridges the audiological and CI literatures.

In people with auditory neuropathy spectrum disorder, who have intact cochlear mechanics but disordered neural timing, the test presented here would reveal whether the displacement from NH reflects the same peripheral geometry as in sensorineural CI users or a qualitatively different pattern.

Preprint – Not peer-reviewed

Online Testing and Large-Scale Normative Data. The most immediate practical extension of this work is an online implementation of the current paradigm, in addition with a version with the proposed modifications.

A standardised stimulus set - built around fixed log-spaced cochlear frequency positions rather than participant-specific electrode centre frequencies - would enable cross-participant and cross-device comparison without normalisation, and would remove the need to enter individual MAP data prior to testing, and even speed up the testing time.

We are currently developing such a platform, which would administer the task remotely, compute the metrics in real time, and benchmark each user's distribution against a normative group of NH reference data collected across age groups and languages.

This would transform the paradigm from a laboratory research tool into a scalable clinical and epidemiological instrument, enabling longitudinal monitoring of perceptual trajectories in large cohorts and the construction of population-level norms against which individual CI users could be evaluated.

Preprint - not peer-reviewed

Bibliography

Abdulrahman Alahmadi, Fida Alzhrani, Abdulrahman Hagr, and Fahad Alghamdi. Pediatric cochlear implantation: The impact of frequency-to-place mismatch on auditory and speech outcomes. *PLOS ONE*, 20(5):e0333088, 2025. doi: 10.1371/journal.pone.0333088.

George Alexiades, Anandhan Dhanasingh, and Claude Jolly. Method to estimate the complete and two-turn cochlear duct length. *Otology & Neurotology*, 36(5):904–907, 2015. doi: 10.1097/MAO.0000000000000620.

Umut Arslan, Hannah Isber, Barbara Shinn-Cunningham, and Antje Ihlefeld. Low-frequency envelope information contributes more than fine structure to speech intelligibility. *Journal of the Acoustical Society of America*, 151(5):3123–3133, 2022. doi: 10.1121/10.0010474.

Deniz Baskent and Robert V. Shannon. Speech recognition under conditions of frequency-place compression and expansion. *The Journal of the Acoustical Society of America*, 113(4):2064–2076, 2004. doi: 10.1121/1.1558357.

Deniz Başkent, Etienne Gaudrain, Terrin N. Tamati, and Anita Wagner. Perception and psychoacoustics of speech in cochlear implant users. In Anthony T. Cacace, Emile de Kleine, Avril Genene Holt, and Pim van Dijk, editors, *Scientific Foundations of Audiology: Perspectives from Physics, Biology, Modeling, and Medicine*, pages 285–319. Plural Publishing, San Diego, CA, 2016. ISBN 978-1-59756-652-0.

Katelyn A. Berg, Hugh M. Birky, Victoria A. Sevich, Aaron C. Moberly, and Terrin N. Tamati. Sound quality, not speech recognition, explains cochlear implant-related quality of life outcomes. *JASA Express Letters*, 5(10):104401, 2025a. doi: 10.1121/10.0039069.

Katelyn A. Berg, Jack H. Noble, Benoit M. Dawant, Leah W. Sunderhaus, Andrea J. DeFreese, Robert F. Labadie, and René H. Gifford. Cochlear implant electrode placement and music perception. *JAMA Otolaryngology–Head & Neck Surgery*, 151(3):220–227, 2025b. doi: 10.1001/jamaoto.2024.4761.

Katrien Berg, Ishan Bhatt, Sona Bhatt, and Bryan E. Pfingst. Selective deactivation of apical electrodes in cochlear implant users. *Trends in Hearing*, 29:1–18, 2024. doi: 10.1177/23312165241298444.

Julie A. Bierer. Threshold and channel interaction in cochlear implant users: evaluation of the tripolar electrode configuration. *The Journal of the Acoustical Society of America*, 121(3):1642–1653, 2007. doi: 10.1121/1.2436712.

Julie A. Bierer and Leonid Litvak. Reducing channel interaction through cochlear implant programming may improve speech perception. *Trends in Hearing*, 20:1–12, 2016. doi: 10.1177/2331216516653389.

Preprint – not peer-reviewed

Julie Arenberg Bierer. Probing the electrode-neuron interface with focused cochlear implant stimulation. *Trends in Amplification*, 14(2):84–95, 2010. doi: 10.1177/1084713810375249.

Julie Arenberg Bierer and Alexandra D. Nye. Comparisons between detection threshold and loudness perception for individual cochlear implant channels. *Ear and Hearing*, 35 (6):641–651, 2014. doi: 10.1097/AUD.0000000000000058.

Robert C. Bilger, F. Owen Black, Neil T. Hopkinson, Eugene N. Myers, James L. Payne, Noel R. Stenson, Armand Vega, and Robert V. Wolf. Evaluation of subjects presently fitted with implanted auditory prostheses. *Annals of Otology, Rhinology & Laryngology*, 86(Suppl 38):1–176, 1977.

Peter Blamey, Françoise Artieres, Deniz Başkent, François Bergeron, Andy Beynon, Emily Burke, Norbert Dillier, Richard Dowell, Bernard Fraysse, Stéphane Gallégo, et al. Factors affecting auditory performance of postlinguistically deaf adults using cochlear implants: An update with 2251 patients. *Audiology and Neurotology*, 18(1):36–47, 2013a.

Peter Blamey et al. Factors affecting auditory performance of postlinguistically deaf adults using cochlear implants: an update with 2251 patients. *Audiology and Neurotology*, 18 (1):36–47, 2013b. doi: 10.1159/000343189.

Colette Boëx, Maria-Izabel Kos, and Marco Pelizzone. Forward masking in different cochlear implant systems. *The Journal of the Acoustical Society of America*, 114(4): 2058–2065, 2003. doi: 10.1121/1.1610452.

Isabelle Boisvert, Maryam Reis, Anna Au, Robert Cowan, and Richard Dowell. Cochlear implantation outcomes in adults: a scoping review. *PLoS ONE*, 15(5):e0232421, 2020a. doi: 10.1371/journal.pone.0232421.

Isabelle Boisvert, Mridula Reis, Andrea Au, Robert Cowan, and Richard C. Dowell. Cochlear implantation outcomes in adults: A scoping review. *PLOS ONE*, 15(5): e0232421, 2020b.

Thomas Bräcker, Markus Meis, Dirk Dittberner, Tobias Wesarg, Antje Aschendorff, and Roland Laszig. Speech intelligibility with a new fine structure coding strategy for the Med-EL cochlear implant system in adults with residual low-frequency hearing. *HNO*, 57(12):1265–1274, 2009. doi: 10.1007/s00106-009-2024-9.

Louis D. Braida, Nathaniel I. Durlach, Richard P. Lippmann, Betty L. Hicks, William M. Rabinowitz, and Charlotte M. Reed. Hearing aids – a review of past research on linear amplification, amplitude compression, and frequency lowering. *ASHA Monographs*, 19: 1–148, 1979.

Jeroen J. Briaire and Johan H. M. Frijns. Unraveling the electrically evoked compound action potential. *Hearing Research*, 205(1–2):143–156, 2005. doi: 10.1016/j.heares.2005.03.020.

Tim Brochier, François Guérit, John M. Deeks, Charlotte Garcia, Manohar Bance, and Robert P. Carlyon. Evaluating and comparing behavioural and electrophysiological estimates of neural health in cochlear implant users. *Journal of the Association for Research in Otolaryngology*, 22(1):67–80, 2021. doi: 10.1007/s10162-020-00773-0.

Preprint – not peer-reviewed

Carolyn J. Brown, Michelle L. Hughes, Barbara Luk, Paul J. Abbas, Amy Wolaver, and Julie Gervais. The relationship between EAP and EABR thresholds and levels used to program the Nucleus 24 speech processor: data from adults. *Ear and Hearing*, 21(2): 151–163, 2000. doi: 10.1097/00003446-200004000-00006.

Nia S. Brown, Sanjena Venkatesh, Tazheh A. Kavoosi, James O. Onyeukwu, Jason A. Brant, and Alexandra E. Quimby. Speech outcomes of cochlear implantation, from 1983 to present: A systematic review. *Otology & Neurotology*, 46(4):393–404, 2025. doi: 10.1097/MAO.0000000000004468.

Lena Bruns, Dirk Mürbe, and Anja Hahne. Understanding music with cochlear implants. *Scientific Reports*, 6:32026, 2016. doi: 10.1038/srep32026.

Meredith T. Caldwell, Nicole T. Jiam, and Charles J. Limb. Assessment and improvement of sound quality in cochlear implant users. *Laryngoscope Investigative Otolaryngology*, 2(3):119–124, 2017a. doi: 10.1002/lio2.71.

Meredith T. Caldwell, Nicole T. Jiam, and Charles J. Limb. Assessment and improvement of sound quality in cochlear implant users. *Laryngoscope Investigative Otolaryngology*, 2(3):119–124, 2017b. doi: 10.1002/lio2.71.

Michael W. Canfarotta, Margaret T. Dillon, Emily Buss, Harold C. Pillsbury, Kevin D. Brown, and Brendan P. O'Connell. Frequency-to-place mismatch: Characterizing variability and the influence on speech perception outcomes in cochlear implant recipients. *Ear and Hearing*, 41(5):1349–1361, 2020a. doi: 10.1097/AUD.0000000000000864.

Michael W. Canfarotta, Margaret T. Dillon, Emily Buss, Harold C. Pillsbury, Kevin D. Brown, and Brendan P. O'Connell. Frequency-to-Place Mismatch: Characterizing Variability and the Influence on Speech Perception Outcomes in Cochlear Implant Recipients. *Ear and Hearing*, 41(5):1349–1361, October 2020b. ISSN 1538-4667. doi: 10.1097/AUD.0000000000000864. URL https://journals.lww.com/ear-hearing/Fulltext/2020/09000/Frequency_to_Place_Mismatch__Characterizing.27.aspx.

Matthew L. Carlson, Douglas P. Sladen, Richard K. Gurgel, Nicole M. Tombers, Christine M. Lohse, and Colin L. Driscoll. Survey of the american neurotology society on cochlear implantation: Part 1, candidacy assessment and expanding indications. *Otology & Neurotology*, 39(1):e12–e19, 2018.

Robert P. Carlyon and Tobias Goehring. Cochlear implant research and development in the twenty-first century: A critical update. *Journal of the Association for Research in Otolaryngology*, 22:481–508, 2021.

Graeme M. Clark. The multiple-channel cochlear implant: the interface between sound and the central nervous system for hearing, speech, and language in deaf people—a personal perspective. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 361:791–810, 2006.

Leah T. Cohen, Lois M. Richardson, Elizabeth Saunders, and Robert S. C. Cowan. Spatial spread of neural excitation in cochlear implant recipients: Comparison of improved ECAP method and psychophysical forward masking. *Hearing Research*, 179(1–2):72–87, 2003. doi: 10.1016/S0378-5955(03)00096-0.

Preprint – not peer-reviewed

Leah T. Cohen, Elizabeth Saunders, Matthew R. Knight, and Robert S. C. Cowan. Psychophysical measures in patients fitted with Contour and straight Nucleus electrode arrays. *Hearing Research*, 212(1–2):160–175, 2006. doi: 10.1016/j.heares.2005.11.005.

Huw R. Cooper, Marilyn Tait, Robert Carpenter, and Thomas P. Nikolopoulos. Spread of excitation measurements in cochlear implant recipients: clinical applications and limitations. In *10th International Conference on Cochlear Implants and Other Implantable Auditory Technologies*, San Diego, CA, 2008. Conference presentation. See also manufacturer-integrated SOE protocols in Cochlear NRT software.

Jason Cramer, Ho-Hsiang Wu, Justin Salamon, and Juan Pablo Bello. Look, listen, and learn more: design choices for deep audio embeddings. pages 3852–3856, 2019. doi: 10.1109/ICASSP.2019.8682475.

Gwenaelle Creff, Nicolas Bernard-Le Liboux, Paul Coudert, Hermine Bourdon, Vincent Pean, Nicolas Wallaert, Cassandre Lambert, and Benoit Godey. Tonotopic and default frequency fitting for music perception in cochlear implant recipients: A randomized clinical trial. *JAMA Otolaryngology–Head & Neck Surgery*, 150(11):960–968, 2024. doi: 10.1001/jamaoto.2024.2895.

Naomi B. H. Croghan, Sarah I. Duran, and Zachary M. Smith. Re-examining the relationship between number of cochlear implant channels and maximal speech intelligibility. *The Journal of the Acoustical Society of America*, 142(6):EL537–EL543, 2017. doi: 10.1121/1.5016044.

Alison DeFreese, Kendra Berg, Jack Noble, Benoit Dawant, Jourdan Holder, et al. Frequency-to-place mismatch and cochlear implant outcomes by electrode type. *JAMA Otolaryngology–Head & Neck Surgery*, 151(2):116–124, 2025. doi: 10.1001/jamaoto.2024.4136.

Andrea DeFreese, Eve Kann, J. Thomas Roland, Mario A. Svirsky, and David M. Landsberger. Frequency-to-place mismatch and speech recognition in cochlear implant users. *Ear & Hearing*, 45(2):330–341, 2024. doi: 10.1097/AUD.0000000000001438.

Ayşenur Karaman Demirel, Ahmet Alperen Akbulut, Ayşe Ayça Çiprut, and Nilüfer Bal. Exploring the link between sound quality perception, music perception, music engagement, and quality of life in cochlear implant recipients. *Audiology Research*, 15 (4):94, 2025. doi: 10.3390/audiolres15040094.

Laura Dessard, Guillaume Gersdorff, Nicola Ivanovik, Masoud Zoca-Assadi, Peter Nopp, Séverine Camby, and Philippe P. Lefebvre. Cochlear implant: analysis of the frequency-to-place mismatch with the table-based software OTOPLAN® and its influence on hearing performance. *Audiology and Neurotology*, 29(3):239–245, 2024.

Lindsay DeVries and Jeroen G. Arenberg. Current focusing to reduce channel interaction for distant electrodes in cochlear implant programs. *Trends in Hearing*, 22: 2331216518813811, 2018. doi: 10.1177/2331216518813811.

Anandhan Dhanasingh and Claude Jolly. An overview of cochlear implant electrode array designs. *Hearing Research*, 356:93–103, 2017. doi: 10.1016/j.heares.2017.10.005.

Preprint - not peer-reviewed

Norbert Dillier, Wai Kong Lai, Brigitta Almqvist, Carolin Frohne, Joachim Müller-Deile, Martin Stecker, and Ernst von Wallenberg. Measurement of the electrically evoked compound action potential via a neural response telemetry system. *Annals of Otology, Rhinology & Laryngology*, 111(5):407–414, 2002. doi: 10.1177/000348940211100505.

Margaret T. Dillon, Michael W. Canfarotta, Emily Buss, and Benjamin P. O'Connell. Comparison of speech recognition with an organ of Corti versus spiral ganglion frequency-to-place function in place-based mapping of cochlear implant and electric-acoustic stimulation devices. *Otology & Neurotology*, 42(5):721–725, 2021. doi: 10.1097/MAO.0000000000003047.

Mishaela DiNino and Julie G. Arenberg. Effects of age and cochlear implantation on spectrally cued speech categorization. *Journal of Speech, Language, and Hearing Research*, 63(8):2425–2440, 2020. doi: 10.1044/2020_JSLHR-19-00127.

André Djourno and Charles Eyriès. Prothèse auditive par excitation électrique à distance du nerf sensoriel à l'aide d'un bobinage inclus à demeure. *La Presse Médicale*, 65:1417, 1957.

Michael F. Dorman et al. Cochlear implant sound quality. *Journal of Speech, Language, and Hearing Research*, 2025. doi: 10.1044/2024_JSLHR-23-00678. TODO: Complete author list and volume/page information.

Ward R Drennan and Jay T Rubinstein. Music perception in cochlear implant users and its relationship with psychophysical capabilities. *Journal of Rehabilitation Research & Development*, 45(5):779–789, 2008. doi: 10.1682/JRRD.2007.08.0118.

Ward R Drennan, Jacob J Oleson, Kate Gfeller, Jillian Crosson, Virginia D Driscoll, Jong Ho Won, Elizabeth S Anderson, and Jay T Rubinstein. Clinical evaluation of music perception, appraisal and experience in cochlear implant users. *International Journal of Audiology*, 54(2):114–123, 2015. doi: 10.3109/14992027.2014.948219.

Michael O. Ernst and Martin S. Banks. Humans integrate visual and haptic information in a statistically optimal fashion. *Nature*, 415(6870):429–433, 2002. doi: 10.1038/415429a.

Andrew Faulkner. Potential benefits from deeply inserted cochlear implant electrodes. *International Journal of Audiology*, 50(S2):S52–S59, 2011. doi: 10.3109/14992027.2010.533704.

Jose N. Fayad and Fred H. Linthicum, Jr. Multichannel cochlear implants: relation of histopathology to performance. *The Laryngoscope*, 116(8):1310–1320, 2006. doi: 10.1097/01.mlg.0000227176.09500.28.

C. C. Finley, T. A. Holden, L. K. Holden, B. R. Whiting, R. A. Chole, G. J. Neely, T. E. Hullar, and M. W. Skinner. Role of electrode placement as a contributor to variability in cochlear implant outcomes. *Otology & Neurotology*, 29(7):920–928, 2008. doi: 10.1097/MAO.0b013e318184f492.

Jill B. Firszt, Laura K. Holden, Margaret W. Skinner, Emily A. Tobey, Amy Peterson, Wolfgang Gaggl, Christina L. Runge-Samuelson, and P. Ashley Wackym. Recognition of speech presented at soft to loud levels by adult cochlear implant recipients of three cochlear implant systems. *Ear & Hearing*, 25(4):375–387, 2004. doi: 10.1097/01.AUD.0000134552.22205.AA.

Preprint - not peer-reviewed

Tom Francart, Jan Brokx, and Jan Wouters. Sensitivity to interaural time differences with combined cochlear implant and acoustic stimulation. *Journal of the Association for Research in Otolaryngology*, 12(6):733–745, 2011. doi: 10.1007/s10162-011-0289-0.

Lindsey M Friesen, Robert V Shannon, Deniz Baskent, and Xiaosong Wang. Speech recognition in noise as a function of the number of spectral channels: Comparison of acoustic hearing and cochlear implants. *Journal of the Acoustical Society of America*, 110(2):1150–1163, 2001. doi: 10.1121/1.1381538.

Johan H. M. Frijns, Jeroen J. Briaire, Jan A. P. M. de Laat, and J. J. Grote. Initial evaluation of the Clarion CII cochlear implant: speech perception and neural response imaging. *Ear and Hearing*, 23(3):184–197, 2002. See also Briaire & Frijns (2005) for spread-of-excitation methodology.

Qian-Jie Fu and John J. Galvin. Perceptual learning following changes in the frequency-to-electrode assignment with the Nucleus-22 cochlear implant. *The Journal of the Acoustical Society of America*, 113(1):351–357, 2003. doi: 10.1121/1.1526428.

Qian-Jie Fu and John J. Galvin. Perceptual learning and auditory training in cochlear implant recipients. *Trends in Amplification*, 11(3):193–205, 2007a. doi: 10.1177/1084713807301379.

Qian-Jie Fu and John J Galvin. Perceptual learning and auditory training in cochlear implant recipients. *Trends in Amplification*, 11(4):193–205, 2007b. doi: 10.1177/1084713807301379.

Qian-Jie Fu and John J. Galvin. Perceptual learning and auditory training in cochlear implant recipients. *Trends in Amplification*, 11(3):193–205, 2007c. doi: 10.1177/1084713807301379.

Christina Fuller, Deniz Başkent, and Rolien Free. Early deafened, late implanted cochlear implant users appreciate music more than and identify music as well as postlingual users. *Frontiers in Neuroscience*, 13:1050, 2019. doi: 10.3389/fnins.2019.01050.

Christopher D Fuller, John J Galvin, Bert Maat, Deniz Båskent, and Rolien H Free. Comparison of two music training approaches on music and speech perception in cochlear implant users. *Trends in Hearing*, 22:2331216518765379, 2018. doi: 10.1177/2331216518765379.

John J. Galvin, Qian-Jie Fu, and Geraldine Nogaki. Melodic contour identification by cochlear implant listeners. *Ear and Hearing*, 28(3):302–319, 2007. doi: 10.1097/01.aud.0000261689.35445.20.

John J Galvin, Elizabeth Eskridge, and Qian-Jie Fu. Deeper cochlear implant electrode insertion angle improves detection of musical sound quality deterioration related to bass frequency removal. *Otology & Neurotology*, 37(2):e92–e98, 2016. doi: 10.1097/MAO.0000000000000941. CHECK: verify author list and page range.

Bruce J. Gantz, Gregg G. Woodworth, John F. Knutson, Paul J. Abbas, and Richard S. Tyler. Multivariate predictors of audiological success with multichannel cochlear implants. *Annals of Otology, Rhinology & Laryngology*, 102(12):909–916, 1993. doi: 10.1177/000348949310201201.

Preprint – not peer-reviewed

Charlotte Garcia, Tobias Goehring, Stefano Cosentino, Richard E. Turner, John M. Deeks, Tim Brochier, Taren Rughooputh, Manohar Bance, and Robert P. Carlyon. The panoramic ECAP method: Estimating patient-specific patterns of current spread and neural health in cochlear implant users. *Journal of the Association for Research in Otolaryngology*, 22(5):567–589, 2021. doi: 10.1007/s10162-021-00795-2.

Ann E. Geers, Christine M. Mitchell, Andrea Warner-Czyz, Nae-Yuh Wang, and Laurie S. Eisenberg. Early sign language exposure and cochlear implantation benefits. *Pediatrics*, 140(1):e20163489, 2017. doi: 10.1542/peds.2016-3489.

Kate Gfeller, Christopher Turner, Jacob Oleson, Xiuhua Zhang, Bruce Gantz, Robin Froman, and Carol Olszewski. Accuracy of cochlear implant recipients on pitch perception, melody recognition, and speech reception in noise. *Ear and Hearing*, 28(3):412–423, 2007. doi: 10.1097/AUD.0b013e3180479318. CHECK: confirm this is the intended paper.

Kate Gfeller, Jacob Oleson, John F Knutson, Patrick Breheny, Virginia Driscoll, and Carol Olszewski. Multivariate predictors of music perception and appraisal by adult cochlear implant users. *Journal of the American Academy of Audiology*, 19(2):120–134, 2008. doi: 10.3766/jaaa.19.2.3.

Kate Gfeller, Dingfeng Jiang, Jacob J. Oleson, Virginia Driscoll, and John F. Knutson. Temporal stability of music perception and appraisal scores of adult cochlear implant recipients. *Journal of the American Academy of Audiology*, 21(1):28–34, 2010. doi: 10.3766/jaaa.21.1.4.

Marcel R. Giezen, Paola Escudero, and Anne Baker. Use of acoustic cues by children with cochlear implants. *Journal of Speech, Language, and Hearing Research*, 53(6): 1440–1457, 2010. doi: 10.1044/1092-4388(2010/09-0107).

René H. Gifford, Michael F. Dorman, Jon K. Shallop, and Sarah A. Sydlowski. Evidence for the expansion of adult cochlear implant candidacy. *Ear and Hearing*, 31(2):186–194, 2010.

René H. Gifford, Jourdan T. Holder, James J. Grisel, Elizabeth L. Perkins, and Robert F. Labadie. Insertion depth and cochlear implant speech recognition outcomes: A comparative study of 28- and 31.5-mm lateral wall arrays. *Otology & Neurotology*, 43(2): e191–e197, 2022. doi: 10.1097/MAO.0000000000003397.

Melanie L Gilbert, Charles J Limb, Eunice K Kim, Mickael L D Deroche, and Rebecca M Lewis. Perception of extreme musical dissonance in cochlear implant users using a novel listening task. *Frontiers in Audiology and Otology*, 3:1554784, 2025. doi: 10.3389/fauot.2025.1554784.

Tobias Goehring, Alexander W Archer-Boyd, John M Deeks, Julie G Arenberg, and Robert P Carlyon. The effect of increased channel interaction on speech perception with cochlear implants. *Scientific Reports*, 11:10424, 2021. doi: 10.1038/s41598-021-89932-8.

Benjamin Goudey, Kerrie Plant, Isabell Kiral, Antonio Jimeno-Yepes, Annalisa Swan, Manoj Gambhir, Andreas Büchner, Eugen Kludt, Robert H. Eikelboom, Cathy Sucher, René H. Gifford, Riaan Rottier, and Hamideh Anjomshoa. A multicenter analysis of

Preprint - not peer-reviewed

factors associated with hearing outcome for 2,735 adults with cochlear implants. *Trends in Hearing*, 25:23312165211037525, 2021. doi: 10.1177/23312165211037525.

Matthew J. Goupell, Christine Stoelb, Alan Kan, and Ruth Y. Litovsky. Effect of mismatched place-of-stimulation on the salience of binaural cues in conditions that simulate bilateral cochlear-implant listening. *The Journal of the Acoustical Society of America*, 133(4):2272–2287, 2013. doi: 10.1121/1.4792936.

Matthew J. Goupell, Olga A. Stakhovskaya, and Joshua G. W. Bernstein. Contralateral interference caused by binaurally presented competing speech in adult bilateral cochlear-implant users. *Ear and Hearing*, 38(1):e110–e123, 2016. doi: 10.1097/AUD.0000000000000357.

Matthew J. Goupell, Stefano Cosentino, Olga A. Stakhovskaya, and Joshua G. W. Bernstein. Interaural pitch-discrimination range effects for bilateral and single-sided-deafness cochlear-implant users. *Journal of the Association for Research in Otolaryngology*, 20 (2):187–203, 2019. doi: 10.1007/s10162-018-00707-x.

Donald D. Greenwood. A cochlear frequency-position function for several species — 29 years later. *The Journal of the Acoustical Society of America*, 87(6):2592–2605, 1990. doi: 10.1121/1.399052.

James D. Harnsberger, Mario A. Svirsky, Adam R. Kaiser, David B. Pisoni, Richard Wright, and Ted A. Meyer. Perceptual "vowel spaces" of cochlear implant users: implications for the study of auditory adaptation to spectral shift. *Journal of the Acoustical Society of America*, 109(5):2135–2145, 2001.

Shawn Harrison, Mehdi Abouzari, Jordan Pang, and Fan-Gang Zeng. Electrocochleography and cognition are important predictors of speech perception outcomes in noise for cochlear implant recipients. *Scientific Reports*, 12:3083, 2022. doi: 10.1038/s41598-022-07175-7.

Shuman He, Jeffrey Skidmore, Brittany Koch, Monita Chatterjee, Brittany L. Carter, and Yi Yuan. The relationship between cochlear implant speech perception outcomes and electrophysiological measures of the electrically evoked compound action potential. *Ear and Hearing*, 45(2):371–384, 2024. doi: 10.1097/AUD.0000000000001289.

James M. Hillenbrand, Laura A. Getty, Michael J. Clark, and Kimberlee Wheeler. Acoustic characteristics of American English vowels. *The Journal of the Acoustical Society of America*, 97(5):3099–3111, 1995. doi: 10.1121/1.411872.

Raul Hinojosa and Margaret Marion. Histopathology of profound sensorineural deafness. *Annals of the New York Academy of Sciences*, 405:459–484, 1983. doi: 10.1111/j.1749-6632.1983.tb31662.x.

Anke Hirschfelder, Silke Gräbel, and Heidi Olze. The impact of cochlear implantation on quality of life: The role of audiologic performance and variables. *Otolaryngology–Head and Neck Surgery*, 138(3):357–362, 2008. doi: 10.1016/j.otohns.2007.10.019.

I. J. Hochmair-Desoyer and E. S. Hochmair. An eight channel scala tympani electrode for auditory prostheses. *IEEE Transactions on Biomedical Engineering*, BME-27(1): 44–50, 1980. doi: 10.1109/TBME.1980.326854.

Preprint – not peer-reviewed

Laura K. Holden, Charles C. Finley, Jill B. Firszt, Timothy A. Holden, Christine Brenner, Lisa G. Potts, Beverly D. Gotter, Shelly S. Vanderhoof, Karen Mispagel, Gina Heydebrand, and Margaret W. Skinner. Factors affecting open-set word recognition in adults with cochlear implants. *Ear and Hearing*, 34(3):342–360, 2013a. doi: 10.1097/AUD.0b013e3182741AA0.

Laura K Holden, Charles C Finley, Jill B Firszt, et al. Factors affecting open-set word recognition in adults with cochlear implants. *Ear and Hearing*, 34(3):342–360, 2013b. doi: 10.1097/AUD.0b013e3182741aa7.

William F. House and Jack Urban. Long term results of electrode implantation and electronic stimulation of the cochlea in man. *Annals of Otology, Rhinology & Laryngology*, 82(4):504–517, 1973. doi: 10.1177/000348947308200408.

Michael J. Hove, Céline Marie, Ian C. Bruce, and Laurel J. Trainor. Superior time perception for lower musical pitch explains why bass-ranged instruments lay down musical rhythms. *Proceedings of the National Academy of Sciences*, 111(28):10383–10388, 2014. doi: 10.1073/pnas.1402039111.

Sarah E. Hughes, Holly A. Hutchings, Frances L. Rapport, Catherine M. McMahon, and Isabelle Boisvert. Social connectedness and perceived listening effort in adult cochlear implant users: A grounded theory to establish content validity for a new patient-reported outcome measure. *Ear and Hearing*, 39(5):922–934, 2018. doi: 10.1097/AUD.0000000000000553.

David Huron. *Sweet Anticipation: Music and the Psychology of Expectation*. MIT Press, Cambridge, MA, 2006. ISBN 9780262083454.

Krista L. Hyde and Isabelle Peretz. Brains that are out of tune but in time. *Psychological Science*, 15(5):356–360, 2004. doi: 10.1111/j.0956-7976.2004.00683.x.

ITU-R – Radiocommunication Sector of ITU. BS.1534-3: Method for the subjective assessment of intermediate quality level of audio systems. Recommendation BS.1534-3, International Telecommunication Union, 2014.

Yannick Jadoul, Bill Thompson, and Bart de Boer. Introducing Parselmouth: A Python interface to Praat. *Journal of Phonetics*, 71:1–15, 2018. doi: 10.1016/j.wocn.2018.07.001.

Christopher J. James, Cyrine Karoui, Marie-Lise Laborde, Benoit Lepage, Corinne Molinier, Joël Tartari, Olivier Deguine, and Bernard Fraysse. Early sentence recognition in adult cochlear implant users. *Ear and Hearing*, 40(4):905–917, 2019. doi: 10.1097/AUD.0000000000000666.

Nicole T. Jiam, Melanie Gilbert, Daniel Cooke, Patpong Jiradejvong, Karen Barrett, Meredith Caldwell, and Charles J. Limb. Association between flat-panel computed tomographic imaging-guided place-pitch mapping and speech and pitch perception in cochlear implant users. *JAMA Otolaryngology–Head & Neck Surgery*, 145(2):109–116, 2019. doi: 10.1001/jamaoto.2018.3096.

Carolyn N. Jolly, Francis A. Spelman, and Ben M. Clopton. Quadrupolar stimulation for cochlear prostheses: Modeling and experimental data. *IEEE Transactions on Biomedical Engineering*, 43(8):857–865, 1996. doi: 10.1109/10.508549.

Preprint – not peer-reviewed

Ruben K. Kalkman, Jeroen J. Briaire, and Johan H. M. Frijns. Stimulation strategies and electrode design in computational models of the electrically stimulated cochlea: an overview of existing literature. *Network: Computation in Neural Systems*, 25(1–2): 1–14, 2014. doi: 10.3109/0954898X.2014.884126.

Alan Kan and Ruth Y. Litovsky. Binaural hearing with electrical stimulation. *Hearing Research*, 322:127–137, 2015. doi: 10.1016/j.heares.2014.08.005.

Adrien M. Khan, Ophir Handzel, Barbara J. Burgess, Diane Damian, Donald K. Eddington, and Joseph B. Nadol, Jr. Is word recognition correlated with the number of surviving spiral ganglion cells and electrode insertion depth in human subjects with cochlear implants? *The Laryngoscope*, 115(4):672–677, 2005. doi: 10.1097/01.mlg.0000161335.62139.80.

David C. Knill and Alexandre Pouget. The bayesian brain: the role of uncertainty in neural coding and computation. *Trends in Neurosciences*, 27(12):712–719, 2004. doi: 10.1016/j.tins.2004.10.007.

Anja Kurz, Fabian-Tobias Müller-Graff, Rudolf Hagen, and Kristen Rak. One click is not enough: Anatomy-based fitting in experienced cochlear implant users. *Otology & Neurotology*, 43(10):1176–1180, 2022. doi: 10.1097/MAO.0000000000003702.

Anja Kurz, Daniel Herrmann, Rudolf Hagen, and Kristen Rak. Using anatomy-based fitting to reduce frequency-to-place mismatch in experienced bilateral cochlear implant users: A promising concept. *Journal of Personalized Medicine*, 13(7):1109, 2023. doi: 10.3390/jpm13071109.

Imre Lahdelma and Tuomas Eerola. Register impacts perceptual consonance through roughness and sharpness. *Psychonomic Bulletin & Review*, 29:1700–1710, 2022. doi: 10.3758/s13423-021-02033-5.

Cynthia Lam, Brian C. J. Moore, Marina Salorio-Corbetto, and Deborah A. Vickers. The relationship between hearing experiences, music-listening behaviors, and chord-discrimination abilities for cochlear implant users. *Trends in Hearing*, 26:1–18, 2022. doi: 10.1177/23312165221143902.

Lara J. G. Lambriks, Marije van Hoof, Jeroen J. Briaire, Johan H. M. Frijns, Laura Hollink, Jeroen van Tongeren, Robert J. Stokroos, and Huib J. Theunisse. Individualized frequency fitting based on cochlear anatomy in experienced cochlear implant users. *Frontiers in Neurology*, 14:1168010, 2023a. doi: 10.3389/fneur.2023.1168010.

Lars Lambriks, Marc van Hoof, Joke Debruyne, Miranda Janssen, Janny Hof, Katja Hellingman, Elke Devocht, and Erwin George. Toward neural health measurements for cochlear implantation: The relationship among electrode positioning, the electrically evoked action potential, impedances and behavioral stimulation levels. *Frontiers in Neurology*, 14:1093265, 2023b. doi: 10.3389/fneur.2023.1093265.

Lars J. G. Lambriks, Marc van Hoof, Joke Debruyne, Miranda Janssen, Josef Chalupper, Katja van der Heijden, Jan R. Hof, Katja Hellingman, Elke Devocht, and Erwin George. Imaging-based frequency mapping for cochlear implants – evaluated using a daily randomized controlled trial. *Frontiers in Neuroscience*, 17:1119933, 2023c. doi: 10.3389/fnins.2023.1119933.

Preprint, not peer reviewed

David M. Landsberger, Maja Svrakic, J. Thomas Roland, and Mario Svirsky. The relationship between insertion angles, default frequency allocations, and spiral ganglion place pitch in cochlear implants. *Ear and Hearing*, 36(5):e207–e213, 2015. doi: 10.1097/AUD.0000000000000163.

Luis Lassaletta, Ana Castro, Maria Bastarrica, Rosa Pérez-Mora, Belen Herrán, Laura Sanz, Maria Jesús De Sarriá, and Javier Gavilán. Does music perception have an impact on quality of life following cochlear implantation? *Acta Oto-Laryngologica*, 128 (6):682–686, 2008. doi: 10.1080/00016480701675212.

Luis Lassaletta, Miryam Calvino, Isabel Sánchez-Cuadrado, and Javier Gavilán. Does it make any sense to fit cochlear implants according to the anatomy-based fitting? Our experience with the first series of patients. *Frontiers in Audiology and Otology*, 1: 1298538, 2023. doi: 10.3389/fauot.2023.1298538.

Diane S. Lazard, Christophe Vincent, Frédéric Venail, Paul Van de Heyning, Eric Truy, Olivier Sterkers, Piotr H. Skarzynski, Henryk Skarzynski, Karen Schauwers, Stephen O'Leary, et al. Pre-, per- and postoperative factors affecting performance of postlinguistically deaf adults using cochlear implants: A new conceptual model over time. *PLOS ONE*, 7(11):e48739, 2012.

E. Lehnhardt. Intracochlear placement of cochlear implant electrodes in soft surgery technique. *HNO*, 41(7):356–359, 1993.

Arne Leijon, Martin Dahlquist, and Karolina Smeds. Bayesian analysis of paired-comparison sound quality ratings. *The Journal of the Acoustical Society of America*, 146(5):3174–3183, 2019. doi: 10.1121/1.5131024.

Rebecca M Lewis, Melanie L Gilbert, Jordan A Beim, Andrew J Oxenham, and Charles J Limb. Using musical pitch interval comparisons to assess cochlear implant frequency-to-place maps. *Frontiers in Audiology and Otology*, 3:1565883, 2025. doi: 10.3389/fauot.2025.1565883.

Michael J. Lichtenstein, Fred H. Bess, and Shirley A. Logan. Diagnostic performance of the hearing handicap inventory for the elderly (screening version) against differing definitions of hearing loss. *Ear & Hearing*, 9(4):208–211, 1988. doi: 10.1097/00003446-198808000-00006.

Jason Tzu-Hsien Lien, Ben Williges, and Deborah Vickers. Cochlear implant remapping informed by measures of viability of the electrode-neural interface: a systematic review with meta-analysis. *Scientific Reports*, 15:09610, 2025. doi: 10.1038/s41598-025-09610-x.

Charles J. Limb and Jay T. Rubinstein. Current research on music perception in cochlear implant users. *Otolaryngologic Clinics of North America*, 45(1):129–140, 2012. doi: 10.1016/j.otc.2011.08.021.

Ruth Y. Litovsky. Spatial release from masking. *Acoustics Today*, 8(2):18–25, 2011.

Ruth Y. Litovsky, Peggy M. Johnstone, and Shelly P. Godar. Benefits of bilateral cochlear implants and/or hearing aids in children. *International Journal of Audiology*, 45(sup1): 78–91, 2006.

Preprint – not peer-reviewed

Ruth Y. Litovsky, Aaron Parkinson, and Jennifer Arcaroli. Spatial hearing and speech intelligibility in bilateral cochlear implant users. *Ear & Hearing*, 30(4):419–431, 2009. doi: 10.1097/AUD.0b013e3181a165be.

Gerald E. Loeb, Carolyn L. Byers, Stephen J. Rebscher, David E. Casey, Michael M. Fong, Robert A. Schindler, Robert F. Gray, and Michael M. Merzenich. Design and fabrication of an experimental cochlear prosthesis. *Medical and Biological Engineering and Computing*, 21(3):241–254, 1983. doi: 10.1007/BF02478489.

Christopher J. Long, Timothy A. Holden, Gary H. McClelland, William S. Parkinson, Clough Shelton, David C. Kelsall, and Zachary M. Smith. Examining the electro-neural interface of cochlear implant users using psychophysics, CT scans, and speech understanding. *Journal of the Association for Research in Otolaryngology*, 15(2):293–304, 2014. doi: 10.1007/s10162-013-0437-5.

Clarinda A. Makary, Jennifer Shin, Sharon G. Kujawa, M. Charles Liberman, and Saumil Bhatt. Age-related primary cochlear neuronal degeneration in human temporal bones. *Journal of the Association for Research in Otolaryngology*, 12(6):711–717, 2011. doi: 10.1007/s10162-011-0283-2.

Rajeev Mathew, Jaime Undurraga, Jingdan Shao, Robert P. Carlyon, Jessica Goodman, and Deborah Vickers. Electrode discrimination and speech perception in newly implanted cochlear implant users. *Hearing Research*, 354:151–163, 2017. doi: 10.1016/j.heares.2017.09.005.

Rajeev Mathew, Deborah Vickers, Patrick Boyle, Azhar Shaida, David Selvadurai, Dan Jiang, and Jaime Undurraga. Development of electrophysiological and behavioural measures of electrode discrimination in adult cochlear implant users. *Hearing Research*, 2018. doi: 10.1016/j.heares.2018.07.002.

Hugh J. McDermott. Music perception with cochlear implants: A review. *Trends in Amplification*, 8(2):49–82, 2004. doi: 10.1177/108471380400800203.

Theodore R. McRackan, Michael Bauschard, Jonathan L. Hatch, Emily Franko-Tobin, H. Ruth Droghini, Shaun A. Nguyen, and Judy R. Dubno. Meta-analysis of quality-of-life improvement after cochlear implantation and associations with speech recognition abilities. *Laryngoscope*, 128(4):982–990, 2018.

Catarina Mendonça and Symeon Delikaris-Manias. Statistical tests with ordinal data in spatial audio research. 2018. Paper 9987.

Jia Meng, Sha Li, Fei Zhang, Qian Li, and Zhao Qin. Cochlear size and shape variability and implications in cochlear implantation surgery. *Otology & Neurotology*, 37(9):1307–1313, 2016. doi: 10.1097/MAO.0000000000001189.

Griet Mertens, Ellen Andries, Andrea Kleine Punte, and Paul Van de Heyning. The smaller the frequency-to-place mismatch the better the hearing outcomes in cochlear implant recipients? *European Archives of Oto-Rhino-Laryngology*, 279(11):5243–5249, 2022. doi: 10.1007/s00405-021-07182-y.

Preprint - not peer-reviewed

Michael M. Merzenich, Robin P. Michelson, C. Robert Pettit, Robert A. Schindler, and Marion Reid. Neural encoding of sound sensation evoked by electrical stimulation of the acoustic nerve. In *Annals of Otology, Rhinology & Laryngology*, volume 82, pages 486–503, 1973. doi: 10.1177/000348947308200407.

Jonathan Mo, Nicole T. Jiam, Mickael L. D. Deroche, Patpong Jiradejvong, and Charles J. Limb. Effect of frequency response manipulations on musical sound quality for cochlear implant users. *Trends in Hearing*, 26:23312165221120017, 2022. doi: 10.1177/23312165221120017.

Aaron C. Moberly, Jonathan H. Lowenstein, Elizabeth Tarr, Amanda Caldwell-Tarr, D. Bradley Welling, Antoine J. Shahin, and Susan Nittrouer. Do adults with cochlear implants rely on different acoustic cues for phoneme perception than adults with normal hearing? *Journal of Speech, Language, and Hearing Research*, 57(2):566–582, 2014. doi: 10.1044/2013_JSLHR-H-13-0013.

Aaron C. Moberly, Jessica H. Lowenstein, and Susan Nittrouer. The enigma of poor performance by adults with cochlear implants. *Otology & Neurotology*, 37(10):1522–1528, 2016.

Brian C. J. Moore. Dead regions in the cochlea: conceptual foundations, diagnosis, and clinical applications. *Ear and Hearing*, 25(2):98–116, 2004. doi: 10.1097/01.AUD.0000120359.49711.D7.

Brian C. J. Moore and Chin-Tuan Tan. Perceived naturalness of spectrally distorted speech and music. *The Journal of the Acoustical Society of America*, 114(1):408–419, 2003. doi: 10.1121/1.1577552.

M. Moran, A. Rousset, and V. Looi. Music appreciation and music listening in prelingual and postlingually deaf adult cochlear implant recipients. *International Journal of Audiology*, 55(Suppl 2):S57–S63, 2016. doi: 10.3109/14992027.2016.1157630.

Isabelle Mosnier, Jean-Pierre Bebear, Mathieu Marx, Bernard Fraysse, Eric Truy, Geneviève Lina-Granade, Michel Mondain, Françoise Sterkers-Artières, Philippe Bordure, Alain Robier, Benoît Godey, Bernard Meyer, Bruno Frachet, Catherine Poncet-Wallet, Didier Bouccara, and Olivier Sterkers. Improvement of cognitive function after cochlear implantation in elderly patients. *JAMA Otolaryngology–Head & Neck Surgery*, 141(5):442–450, 2015. doi: 10.1001/jamaoto.2015.129.

Joseph B. Nadol. Patterns of neural degeneration in the human cochlea and auditory nerve: Implications for cochlear implantation. *Otolaryngology–Head and Neck Surgery*, 117(3):220–228, 1997. doi: 10.1016/S0194-5998(97)70178-5.

Leanne Nagels, Etienne Gaudrain, Deborah Vickers, Petra Hendriks, and Deniz Başkent. Prelingually deaf children with cochlear implants show better perception of voice cues and speech in competing speech than postlingually deaf adults with cochlear implants. *Ear & Hearing*, 45(4):952–968, 2024. doi: 10.1097/AUD.0000000000001489.

National Institute on Deafness and Other Communication Disorders. Cochlear implants, 2024. URL `https://www.nidcd.nih.gov/health/cochlear-implants`. Accessed: 2024.

Preprint – not peer-reviewed

John K. Niparko, Emily A. Tobey, Donna J. Thal, Laurie S. Eisenberg, Nae-Yuh Wang, Alexandra L. Quittner, and Nancy E. Fink. Spoken language development in children following cochlear implantation. *JAMA*, 303(15):1498–1506, 2010. doi: 10.1001/jama.2010.451.

Jack H. Noble, Robert F. Labadie, Renée H. Gifford, and Benoit M. Dawant. Image-guidance enables new methods for customizing cochlear implant stimulation strategies. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 21(5):820–829, 2013. doi: 10.1109/TNSRE.2013.2253333.

Jack H. Noble, René H. Gifford, Andrea J. Hedley-Williams, Benoit M. Dawant, and Robert F. Labadie. Clinical evaluation of an image-guided cochlear implant programming strategy. *Audiology and Neurotology*, 19(3):152–160, 2020.

Sylvie Nozaradan, Isabelle Peretz, Peter E. Keller, Peter Vuust, and André Mouraux. Neural tracking of the musical beat is enhanced by low-frequency sounds. *Proceedings of the National Academy of Sciences*, 115(32):8221–8226, 2018. doi: 10.1073/pnas.1801421115. CHECK: verify full author list.

Heidi Olze, Agnieszka J. Szczepek, Heidemarie Haupt, Ulrike Förster, Nora Zirke, Silke Gräbel, and Birgit Mazurek. Cochlear implantation has a positive influence on quality of life, tinnitus, and psychological comorbidity. *The Laryngoscope*, 121(10):2220–2227, 2011. doi: 10.1002/lary.22145.

Richard Parncutt. Revision of Terhardt's psychoacoustical model of the root(s) of a musical chord. *Music Perception*, 6(1):65–93, 1988. doi: 10.2307/40285416. CHECK doi.

Gordon E Peterson and Harold L Barney. Control methods used in a study of the vowels. *The Journal of the Acoustical Society of America*, 24(2):175–184, 1952. doi: 10.1121/1.1906875.

Nathaniel R. Peterson, David B. Pisoni, and Richard T. Miyamoto. Cochlear implants and spoken language processing abilities: review and assessment of the literature. *Restorative Neurology and Neuroscience*, 28(2):237–250, 2010. doi: 10.3233/RNN-2010-0535.

Bryan E. Pfingst, Deborah J. Colesa, Sara Hembrador, Sung Yun Kang, John C. Middlebrooks, Yehoash Raphael, and Grace L. Su. Detection of pulse trains in the electrically stimulated cochlea: effects of cochlear health. *Journal of the Acoustical Society of America*, 130(6):3954–3968, 2011a. doi: 10.1121/1.3651820.

Bryan E. Pfingst, Deborah J. Colesa, Stefan Hembrador, Su-Wooi Kang, John C. Middlebrooks, Yehoash Raphael, and Gina L. Su. Detection of pulse trains in the electrically stimulated cochlea: Effects of cochlear health. *The Journal of the Acoustical Society of America*, 130(6):3954–3968, 2011b. doi: 10.1121/1.3651820.

Bryan E Pfingst, Ning Zhou, Deborah J Colesa, Margaret M Watts, Stefan B Strahl, Yehoash Raphael, Richard A Altschuler, and Suyash Bhatt. Importance of cochlear health for implant function. *Hearing Research*, 322:77–88, 2015. doi: 10.1016/j.heares.2014.09.009.

Preprint - not peer-reviewed

Reinier Plomp and Willem J. M. Levelt. Tonal consonance and critical bandwidth. *Journal of the Acoustical Society of America*, 38(4):548–560, 1965. doi: 10.1121/1.1909741.

Frank Rattay, Patrick Lutter, and Hubert Felix. A model of the electrically excited human cochlear neuron. I. Contribution of neural substructures to the generation and propagation of spikes. *Hearing Research*, 153(1–2):43–63, 2001. doi: 10.1016/S0378-5955(00)00256-2.

Lina A. J. Reiss, Christopher W. Turner, Sandy R. Erenberg, and Bruce J. Gantz. Changes in pitch with a cochlear implant over time. *Journal of the Association for Research in Otolaryngology*, 8(2):241–257, 2007. doi: 10.1007/s10162-007-0077-8.

Lina A. J. Reiss, Courtney S. Shayman, Emily P. Walker, Keri O. Bennett, Jennifer R. Fowler, Christy L. Hartling, Beth Glickman, Michael R. Lasarev, and Yonghee Oh. Binaural pitch fusion: Comparison of normal-hearing and hearing-impaired listeners. *The Journal of the Acoustical Society of America*, 143(3):1642–1653, 2018. doi: 10.1121/1.5026934.

Sara K Röbler, Blake P O'Connell, Jacob B Hunter, René H Gifford, Colin L W Driscoll, and Matthew L Carlson. Electrocochleography and cognition are important predictors of speech perception outcomes in noise for cochlear implant recipients. *Scientific Reports*, 12:3083, 2022. doi: 10.1038/s41598-022-07175-7.

Alexandre T. Roy, Martin J. Penner, Anuj Bhatt, Vijayalakshmi Rao, Chan-Chan Lau, and Fan-Gang Zeng. Sound quality after cochlear implantation: effects of electrode array position. *Ear & Hearing*, 37(2):e100–e109, 2016. doi: 10.1097/AUD.0000000000000244.

Alexis T. Roy, Patpong Jiradejvong, Courtney Carver, and Charles J. Limb. Assessment of sound quality perception in cochlear implant users during music listening. *Otology & Neurotology*, 33(3):319–327, 2012. doi: 10.1097/MAO.0b013e31824296a9.

Eitan Sagi and Mario A. Svirsky. A level-adjusted cochlear frequency-to-place map for estimating tonotopic frequency mismatch with a cochlear implant. *Ear and Hearing*, 46(4):963–975, 2025. doi: 10.1097/AUD.0000000000001641.

Reinhold Schatzer, Katrien Vermeire, Dimphna Visser, Andreas Krenmayr, Michael Kals, Marc Voormolen, Paul Van de Heyning, and Clemens Zierhofer. Electric-acoustic pitch comparisons in single-sided-deaf cochlear implant users: Frequency-place functions and rate pitch. *Hearing Research*, 309:26–35, 2014. doi: 10.1016/j.heares.2013.11.003.

Kara C. Schvartz-Leyzac and Bryan E. Pfingst. Across-site patterns of electrically evoked compound action potential amplitude-growth functions in multichannel cochlear implant recipients and the effects of the interphase gap. *Hearing Research*, 341:50–65, 2016. doi: 10.1016/j.heares.2016.08.002.

Robert V. Shannon, Fan-Gang Zeng, Vinay Kamath, John Wygonski, and Michael Ekelid. Speech recognition with primarily temporal cues. *Science*, 270(5234):303–304, 1995. doi: 10.1126/science.270.5234.303.

Preprint - not peer-reviewed

Margaret W. Skinner, Timothy A. Holden, Barbara R. Whiting, Arne H. Voie, Barry Brunsden, John G. Neely, Eric A. Saxon, Timothy E. Hullar, and Robert A. Petrel. In vivo estimates of the position of Advanced Bionics electrode arrays in the human cochlea. *Annals of Otology, Rhinology & Laryngology*, 111(9 Suppl):2–24, 2002. doi: 10.1177/00034894021110S901.

Anthony J. Spahr, Michael F. Dorman, Leonid M. Litvak, Sasha Van Wie, Renée H. Gifford, Philipos C. Loizou, Louise M. Loiselle, Terri Oakes, and Sara Cook. Development and validation of the AzBio sentence lists. *Ear and Hearing*, 33(1):112–117, 2012. doi: 10.1097/AUD.0b013e31822c2549.

Aniket G. Srinivasan, Mario Padilla, Robert V. Shannon, and David M. Landsberger. Improving speech perception in noise with current focusing in cochlear implant users. *Hearing Research*, 299:29–36, 2013. doi: 10.1016/j.heares.2013.02.004.

Olga Stakhovskaya, Dhananjay Sridhar, Benson H. Bonham, and Patricia A. Leake. Frequency map for the human cochlear spiral ganglion: implications for cochlear implants. *Journal of the Association for Research in Otolaryngology*, 8(2):220–233, 2007a.

Olga Stakhovskaya, Dhiresha Sridhar, Ben H Bonham, and Patricia A Leake. Frequency map for the human cochlear spiral ganglion: implications for cochlear implants. *Journal of the Association for Research in Otolaryngology*, 8(2):220–233, 2007b. doi: 10.1007/s10162-007-0076-9.

Jacob J. Sturm, Viraj P. Patel, Graham Dibelius, Manfred Kuhlmey, and Ana H. Kim. Frequency-to-place mismatch impacts cochlear implant quality of life, but not speech recognition. *The Laryngoscope*, 134(6):2820–2827, 2024. doi: 10.1002/lary.31264.

Anita Sulibhavi, Christopher Z. Wen, Kevin L. Li, and Tiffany P. Hwa. The impact of cochlear implant experience on music perception: Systematic review and meta-analysis. *Current Otorhinolaryngology Reports*, 10:315–321, 2022. doi: 10.1007/s40136-022-00418-1.

Mario A. Svirsky, Adriana Silveira, Héctor Suarez, Harold Neuburger, Thanh T. Lai, J. B. Simmons, and Anne L. Beiter. Auditory learning and adaptation after cochlear implantation: a preliminary study of discrimination and labelling of vowel sounds by cochlear implant users. *Acta Oto-Laryngologica*, 125(2):144–153, 2005.

Mario A Svirsky, Ana Silveira, Howard Neuburger, Howard Suarez, and Tina T Lai. A model of incomplete adaptation to a severely shifted frequency-to-electrode mapping by cochlear implant users. *Journal of the Association for Research in Otolaryngology*, 11(4):565–579, 2010. doi: 10.1007/s10162-010-0214-x.

Ernst Terhardt. The concept of musical consonance: a link between music and psychoacoustics. *Music Perception*, 1(3):276–295, 1984. doi: 10.2307/40285261. CHECK doi.

The MathWorks Inc. MATLAB r2023a, 2023. URL `https://www.mathworks.com`.

Thilo Thiede, William C. Treurniet, Roland Bitto, Christian Schmidmer, Thomas Sporer, Jannes G. Beerens, Catherine Conseldo, Michael Keyhl, Gerhard Stoll, Karlheinz Brandenburg, and Bernd Feiten. PEAQ – the ITU standard for objective measurement of perceived audio quality. *Journal of the Audio Engineering Society*, 48(1-2):3–29, 2000.

Preprint – not peer-reviewed

Chandra S. Throckmorton and Leslie M. Collins. Investigation of the effects of temporal and spatial interactions on speech-recognition skills in cochlear-implant subjects. *The Journal of the Acoustical Society of America*, 105(2):861–873, 1999. doi: 10.1121/1.426275.

Y. C. Tong, R. C. Black, G. M. Clark, I. C. Forster, R. C. Dowell, L. F. A. Martin, and J. F. Patrick. A preliminary report on a multiple-channel cochlear implant operation. *The Journal of Laryngology and Otology*, 93(7):679–695, 1979. doi: 10.1017/S0022215100087771.

Yit Chow Tong, Graeme M. Clark, Peter J. Blamey, Peter A. Busby, and Richard C. Dowell. Psychophysical studies for two multiple-channel cochlear implant patients. *The Journal of the Acoustical Society of America*, 71(1):153–160, 1982. doi: 10.1121/1.387342.

Frederic Venail, Caroline Mathiolon, Sophie Menjot de Champfleur, Jean Pierre Piron, Marielle Sicard, Françoise Villemus, Marie Aude Vessigaud, Françoise Sterkers-Artieres, Michel Mondain, and Alain Uziel. Effects of Electrode Array Length on Frequency-Place Mismatch and Speech Perception with Cochlear Implants. *Audiology and Neurotology*, 20(2):102–111, 2015a. ISSN 1420-3030, 1421-9700. doi: 10.1159/000369333. URL https://www.karger.com/Article/FullText/369333.

Frédéric Venail, Camille Mathiolon, Sophie Menjot de Champfleur, Jean-Pierre Piron, Stéphanie Siber, Françoise Artieres-Roux, Alain Uziel, and Michel Mondain. Effects of electrode array length on frequency-place mismatch and speech perception with cochlear implants. *Audiology and Neurotology*, 20(3):171–182, 2015b. doi: 10.1159/000369333.

Alessandro Volta. On the electricity excited by the mere contact of conducting substances of different kinds. *Philosophical Transactions of the Royal Society of London*, 90:403–431, 1800.

C. von Ilberg, J. Kiefer, J. Tillein, T. Pfenningdorff, R. Hartmann, E. Stürzebecher, and R. Klinke. Electric-acoustic stimulation of the auditory system: new technology for severe hearing loss. *ORL: Journal for Oto-Rhino-Laryngology and Its Related Specialties*, 61(6):334–340, 1999. doi: 10.1159/000027695.

Amit Walia, Amanda J. Ortmann, Sara Lefler, Timothy A. Holden, Sidharth V. Puram, Jacques A. Herzog, and Craig A. Buchman. Electrocochleography-based tonotopic map: II. Frequency-to-place mismatch impacts speech-perception outcomes in cochlear implant recipients. *Ear and Hearing*, 46(1):265–276, 2025. doi: 10.1097/AUD.0000000000001579.

Dorji Wangchuk, Yoon Ah Kim, Sung Hoon Choi, and Il Joon Moon. Anatomy-based fitting in cochlear implant users: Effect on speech perception and subjective outcomes. *Clinical and Experimental Otorhinolaryngology*, 17(2):161–170, 2024. doi: 10.21053/ceo.2023.01234.

Ben Williges and Tim Jürgens. A flexible vocoder toolbox for acoustic simulations of cochlear implants in MATLAB. *arXiv preprint*, 2019. arXiv:1910.02541.

Preprint - not peer-reviewed

Blake S. Wilson and Michael F. Dorman. Cochlear implants: a remarkable past and a brilliant future. *Hearing Research*, 242(1-2):3–21, 2007. doi: 10.1016/j.heares.2007.11.001.

Blake S. Wilson and Michael F. Dorman. Cochlear implants: A remarkable past and a brilliant future. *Hearing Research*, 242(1–2):3–21, 2008. doi: 10.1016/j.heares.2008.06.005.

Blake S. Wilson, Charles C. Finley, Dewey T. Lawson, Robert D. Wolford, Donald K. Eddington, and William M. Rabinowitz. Better speech recognition with cochlear implants. *Nature*, 352:236–238, 1991.

Ruijie Yang, Ting Xu, Yang Liu, Chun Chen, Zhigang Wang, and Yanmei Li. The effect of anatomically based frequency allocation on speech perception of cochlear implant users: A systematic review and meta-analysis. *Frontiers in Neuroscience*, 16:834036, 2022. doi: 10.3389/fnins.2022.834036.

Diego Zanetti, Maurizio Guida, Giulia Bianchin, Andrea Frisina, and Marco Guglieri. Tonotopic fitting in cochlear implant users: a prospective, double-blind, randomised crossover study. *Acta Oto-Laryngologica*, 141(5):449–455, 2021. doi: 10.1080/00016489.2021.1875462.

Robert J. Zatorre and Valorie N. Salimpoor. From perception to pleasure: Music and its neural substrates. *Proceedings of the National Academy of Sciences*, 110(Suppl. 2):10430–10437, 2013. doi: 10.1073/pnas.1301228110.

Fan-Gang Zeng, Stephen Rebscher, William Harrison, Xiaoan Sun, and Haihong Feng. Cochlear implants: System design, integration, and evaluation. *IEEE Reviews in Biomedical Engineering*, 1:115–142, 2008. doi: 10.1109/RBME.2008.2008250.

Sławomir K. Zieliński, Francis Rumsey, and Søren Bech. On some biases encountered in modern audio quality listening tests — a review. *Journal of the Audio Engineering Society*, 56(6):427–451, 2007.

Vera Zimmer, Jürgen L. Verhey, Michael Ziese, and Maike Böckmann-Barthel. Harmony perception in prelingually deaf, juvenile cochlear implant users. *Frontiers in Neuroscience*, 13:466, 2019. doi: 10.3389/fnins.2019.00466.

Preprint - not peer-reviewed